Filter Summary Report: CG,TIA,Automated,CROSS,COUPLED,some,parasitic,Z1,Z3,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated CROSS COUPLED some parasitic Z1 Z3 ZL: $\frac{Z_1Z_3Z_Lg_mr_o+Z_1Z_3Z_L}{Z_1Z_3g_mr_o+Z_1Z_3+Z_1Z_Lg_mr_o+Z_1Z_L+Z_3Z_L+Z_3r_o+Z_Lr_o}$

$$H(z) = \frac{Z_1Z_3Z_Lg_mr_o + Z_1Z_3Z_L}{Z_1Z_3g_mr_o + Z_1Z_3 + Z_1Z_Lg_mr_o + Z_1Z_L + Z_3Z_L + Z_3r_o + Z_Lr_o}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(R_1,\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{L_Ls}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{s\left(L_LR_1R_3g_mr_o + L_LR_1R_3\right)}{R_1R_3g_mr_o + R_1R_3 + R_3r_o + s^2\left(C_LL_LR_1R_3g_mr_o + C_LL_LR_1R_3 + C_LL_LR_3r_o\right) + s\left(L_LR_1g_mr_o + L_LR_1 + L_LR_3 + L_Lr_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_L}R_1R_3g_mr_o+\sqrt{C_L}R_1R_3+\sqrt{C_L}R_3r_o}{\sqrt{L_L}R_1g_mr_o+\sqrt{L_L}R_1+\sqrt{L_L}R_3+\sqrt{L_L}r_o}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{\sqrt{L_L}R_1g_mr_o+\sqrt{L_L}R_1+\sqrt{L_L}R_3+\sqrt{L_L}r_o}{\sqrt{C_L}\sqrt{L_L}\left(\sqrt{C_L}R_1R_3g_mr_o+\sqrt{C_L}R_1R_3+\sqrt{C_L}R_3r_o\right)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1R_3g_mr_o+R_1R_3}{R_1g_mr_o+R_1+R_3+r_o}$
Qz: None
Wz: None

3.2 BP-2 $Z(s) = \left(R_1,\ \infty,\ \frac{1}{C_3s},\ \infty,\ \infty,\ \frac{L_Ls}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{s\left(L_LR_1g_mr_o + L_LR_1\right)}{L_Ls + R_1g_mr_o + R_1 + r_o + s^2\left(C_3L_LR_1g_mr_o + C_3L_LR_1 + C_3L_Lr_o + C_LL_LR_1g_mr_o + C_LL_LR_1 + C_LL_Lr_o\right)}$$

Parameters:

Q: $\frac{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}{\sqrt{L_L}\sqrt{C_3+C_L}}$
wo: $\frac{1}{\sqrt{C_3L_L+C_LL_L}}$
bandwidth: $\frac{\sqrt{L_L}\sqrt{C_3+C_L}}{\sqrt{C_3L_L+C_LL_L}(C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o)}$
K-LP: 0
K-HP: 0
K-BP: $R_1g_mr_o + R_1$
Qz: None
Wz: None

3.3 BP-3 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ \infty,\ \frac{L_Ls}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{s\left(L_LR_1R_3g_mr_o + L_LR_1R_3\right)}{R_1R_3g_mr_o + R_1R_3 + R_3r_o + s^2\left(C_3L_LR_1R_3g_mr_o + C_3L_LR_1R_3 + C_3L_LR_3r_o + C_LL_LR_1R_3g_mr_o + C_LL_LR_1R_3 + C_LL_LR_3r_o\right) + s\left(L_LR_1g_mr_o + L_LR_1 + L_LR_3 + L_Lr_o\right)}$$

Parameters:

Q: $\frac{C_3R_1R_3g_mr_o+C_3R_1R_3+C_3R_3r_o+C_LR_1R_3g_mr_o+C_LR_1R_3+C_LR_3r_o}{\sqrt{L_L}R_1g_mr_o\sqrt{C_3+C_L}+\sqrt{L_L}R_1\sqrt{C_3+C_L}+\sqrt{L_L}R_3\sqrt{C_3+C_L}+\sqrt{L_L}r_o\sqrt{C_3+C_L}}$
wo: $\frac{1}{\sqrt{C_3L_L+C_LL_L}}$
bandwidth: $\frac{\sqrt{L_L}R_1g_mr_o\sqrt{C_3+C_L}+\sqrt{L_L}R_1\sqrt{C_3+C_L}+\sqrt{L_L}R_3\sqrt{C_3+C_L}+\sqrt{L_L}r_o\sqrt{C_3+C_L}}{\sqrt{C_3L_L+C_LL_L}(C_3R_1R_3g_mr_o+C_3R_1R_3+C_3R_3r_o+C_LR_1R_3g_mr_o+C_LR_1R_3+C_LR_3r_o)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1R_3g_mr_o+R_1R_3}{R_1g_mr_o+R_1+R_3+r_o}$
Qz: None
Wz: None

3.4 BP-4 $Z(s) = \left(R_1,\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s\left(L_3 R_1 R_L g_m r_o + L_3 R_1 R_L\right)}{R_1 R_L g_m r_o + R_1 R_L + R_L r_o + s^2\left(C_3 L_3 R_1 R_L g_m r_o + C_3 L_3 R_1 R_L + C_3 L_3 R_L r_o\right) + s\left(L_3 R_1 g_m r_o + L_3 R_1 + L_3 R_L + L_3 r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_3} R_1 R_L g_m r_o + \sqrt{C_3} R_1 R_L + \sqrt{C_3} R_L r_o}{\sqrt{L_3} R_1 g_m r_o + \sqrt{L_3} R_1 + \sqrt{L_3} R_L + \sqrt{L_3} r_o}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{\sqrt{L_3} R_1 g_m r_o + \sqrt{L_3} R_1 + \sqrt{L_3} R_L + \sqrt{L_3} r_o}{\sqrt{C_3}\sqrt{L_3}\left(\sqrt{C_3} R_1 R_L g_m r_o + \sqrt{C_3} R_1 R_L + \sqrt{C_3} R_L r_o\right)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$
Qz: None
Wz: None

3.5 BP-5 $Z(s) = \left(R_1,\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{s\left(L_3 R_1 g_m r_o + L_3 R_1\right)}{L_3 s + R_1 g_m r_o + R_1 + r_o + s^2\left(C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 r_o + C_L L_3 R_1 g_m r_o + C_L L_3 R_1 + C_L L_3 r_o\right)}$$

Parameters:

Q: $\frac{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}{\sqrt{L_3}\sqrt{C_3 + C_L}}$
wo: $\frac{1}{\sqrt{C_3 L_3 + C_L L_3}}$
bandwidth: $\frac{\sqrt{L_3}\sqrt{C_3 + C_L}}{\sqrt{C_3 L_3 + C_L L_3}\left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right)}$
K-LP: 0
K-HP: 0
K-BP: $R_1 g_m r_o + R_1$
Qz: None
Wz: None

3.6 BP-6 $Z(s) = \left(R_1,\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{s\left(L_3 R_1 R_L g_m r_o + L_3 R_1 R_L\right)}{R_1 R_L g_m r_o + R_1 R_L + R_L r_o + s^2\left(C_3 L_3 R_1 R_L g_m r_o + C_3 L_3 R_1 R_L + C_3 L_3 R_L r_o + C_L L_3 R_1 R_L g_m r_o + C_L L_3 R_1 R_L + C_L L_3 R_L r_o\right) + s\left(L_3 R_1 g_m r_o + L_3 R_1 + L_3 R_L + L_3 r_o\right)}$$

Parameters:

Q: $\frac{C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}{\sqrt{L_3} R_1 g_m r_o \sqrt{C_3 + C_L} + \sqrt{L_3} R_1 \sqrt{C_3 + C_L} + \sqrt{L_3} R_L \sqrt{C_3 + C_L} + \sqrt{L_3} r_o \sqrt{C_3 + C_L}}$
wo: $\frac{1}{\sqrt{C_3 L_3 + C_L L_3}}$
bandwidth: $\frac{\sqrt{L_3} R_1 g_m r_o \sqrt{C_3 + C_L} + \sqrt{L_3} R_1 \sqrt{C_3 + C_L} + \sqrt{L_3} R_L \sqrt{C_3 + C_L} + \sqrt{L_3} r_o \sqrt{C_3 + C_L}}{\sqrt{C_3 L_3 + C_L L_3}\left(C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o\right)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$
Qz: None
Wz: None

3.7 BP-7 $Z(s) = \left(R_1,\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s\left(L_3 L_L R_1 g_m r_o + L_3 L_L R_1\right)}{L_3 L_L s + L_3 R_1 g_m r_o + L_3 R_1 + L_3 r_o + L_L R_1 g_m r_o + L_L R_1 + L_L r_o + s^2\left(C_3 L_3 L_L R_1 g_m r_o + C_3 L_3 L_L R_1 + C_3 L_3 L_L r_o + C_L L_3 L_L R_1 g_m r_o + C_L L_3 L_L R_1 + C_L L_3 L_L r_o\right)}$$

Parameters:

Q: $\frac{C_3 R_1 g_m r_o \sqrt{L_3 + L_L} + C_3 R_1 \sqrt{L_3 + L_L} + C_3 r_o \sqrt{L_3 + L_L} + C_L R_1 g_m r_o \sqrt{L_3 + L_L} + C_L R_1 \sqrt{L_3 + L_L} + C_L r_o \sqrt{L_3 + L_L}}{\sqrt{L_3}\sqrt{L_L}\sqrt{C_3 + C_L}}$
wo: $\frac{\sqrt{L_3 + L_L}}{\sqrt{C_3 L_3 L_L + C_L L_3 L_L}}$

bandwidth: $\frac{\sqrt{L_3}\sqrt{L_L}\sqrt{C_3+C_L}\sqrt{L_3+L_L}}{\sqrt{C_3L_3L_L+C_LL_3L_L}\left(C_3R_1g_mr_o\sqrt{L_3+L_L}+C_3R_1\sqrt{L_3+L_L}+C_3r_o\sqrt{L_3+L_L}+C_LR_1g_mr_o\sqrt{L_3+L_L}+C_LR_1\sqrt{L_3+L_L}+C_Lr_o\sqrt{L_3+L_L}\right)}$
K-LP: 0
K-HP: 0
K-BP: $R_1g_mr_o + R_1$
Qz: None
Wz: None

3.8 BP-8 $Z(s) = \left(\frac{L_1s}{C_1L_1s^2+1},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s\left(L_1R_3R_Lg_mr_o + L_1R_3R_L\right)}{R_3R_L + R_3r_o + R_Lr_o + s^2\left(C_1L_1R_3R_L + C_1L_1R_3r_o + C_1L_1R_Lr_o\right) + s\left(L_1R_3g_mr_o + L_1R_3 + L_1R_Lg_mr_o + L_1R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_3R_L+\sqrt{C_1}R_3r_o+\sqrt{C_1}R_Lr_o}{\sqrt{L_1}R_3g_mr_o+\sqrt{L_1}R_3+\sqrt{L_1}R_Lg_mr_o+\sqrt{L_1}R_L}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_3g_mr_o+\sqrt{L_1}R_3+\sqrt{L_1}R_Lg_mr_o+\sqrt{L_1}R_L}{\sqrt{C_1}\sqrt{L_1}\left(\sqrt{C_1}R_3R_L+\sqrt{C_1}R_3r_o+\sqrt{C_1}R_Lr_o\right)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_3R_L}{R_3+R_L}$
Qz: None
Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(R_1,\ \infty,\ R_3,\ \infty,\ \infty,\ L_Ls + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_1R_3g_mr_o + R_1R_3 + s^2\left(C_LL_LR_1R_3g_mr_o + C_LL_LR_1R_3\right)}{R_1g_mr_o + R_1 + R_3 + r_o + s^2\left(C_LL_LR_1g_mr_o + C_LL_LR_1 + C_LL_LR_3 + C_LL_Lr_o\right) + s\left(C_LR_1R_3g_mr_o + C_LR_1R_3 + C_LR_3r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{L_L}R_1g_mr_o+\sqrt{L_L}R_1+\sqrt{L_L}R_3+\sqrt{L_L}r_o}{\sqrt{C_L}R_1R_3g_mr_o+\sqrt{C_L}R_1R_3+\sqrt{C_L}R_3r_o}$
wo: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$
bandwidth: $\frac{\sqrt{C_L}R_1R_3g_mr_o+\sqrt{C_L}R_1R_3+\sqrt{C_L}R_3r_o}{\sqrt{C_L}\sqrt{L_L}\left(\sqrt{L_L}R_1g_mr_o+\sqrt{L_L}R_1+\sqrt{L_L}R_3+\sqrt{L_L}r_o\right)}$
K-LP: $\frac{R_1R_3g_mr_o+R_1R_3}{R_1g_mr_o+R_1+R_3+r_o}$
K-HP: $\frac{R_1R_3g_mr_o+R_1R_3}{R_1g_mr_o+R_1+R_3+r_o}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L}\sqrt{L_L}}$

4.2 BS-2 $Z(s) = \left(R_1,\ \infty,\ L_3s + \frac{1}{C_3s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1R_Lg_mr_o + R_1R_L + s^2\left(C_3L_3R_1R_Lg_mr_o + C_3L_3R_1R_L\right)}{R_1g_mr_o + R_1 + R_L + r_o + s^2\left(C_3L_3R_1g_mr_o + C_3L_3R_1 + C_3L_3R_L + C_3L_3r_o\right) + s\left(C_3R_1R_Lg_mr_o + C_3R_1R_L + C_3R_Lr_o\right)}$$

Parameters:

Q: $\frac{\sqrt{L_3}R_1g_mr_o+\sqrt{L_3}R_1+\sqrt{L_3}R_L+\sqrt{L_3}r_o}{\sqrt{C_3}R_1R_Lg_mr_o+\sqrt{C_3}R_1R_L+\sqrt{C_3}R_Lr_o}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$
bandwidth: $\frac{\sqrt{C_3}R_1R_Lg_mr_o+\sqrt{C_3}R_1R_L+\sqrt{C_3}R_Lr_o}{\sqrt{C_3}\sqrt{L_3}\left(\sqrt{L_3}R_1g_mr_o+\sqrt{L_3}R_1+\sqrt{L_3}R_L+\sqrt{L_3}r_o\right)}$
K-LP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$
K-HP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$
K-BP: 0
Qz: None

Wz: $\frac{1}{\sqrt{C_3}\sqrt{L_3}}$

4.3 BS-3 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L + s^2 \left(C_1 L_1 R_3 R_L g_m r_o + C_1 L_1 R_3 R_L\right)}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s^2 \left(C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3 + C_1 L_1 R_L g_m r_o + C_1 L_1 R_L\right) + s \left(C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{L_1} R_3 g_m r_o + \sqrt{L_1} R_3 + \sqrt{L_1} R_L g_m r_o + \sqrt{L_1} R_L}{\sqrt{C_1} R_3 R_L + \sqrt{C_1} R_3 r_o + \sqrt{C_1} R_L r_o}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{C_1} R_3 R_L + \sqrt{C_1} R_3 r_o + \sqrt{C_1} R_L r_o}{\sqrt{C_1}\sqrt{L_1}\left(\sqrt{L_1} R_3 g_m r_o + \sqrt{L_1} R_3 + \sqrt{L_1} R_L g_m r_o + \sqrt{L_1} R_L\right)}$
K-LP: $\frac{R_3 R_L}{R_3 + R_L}$
K-HP: $\frac{R_3 R_L}{R_3 + R_L}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3}{C_1 C_L R_3 r_o s^2 + g_m r_o + s \left(C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_3} r_o \sqrt{g_m + \frac{1}{r_o}}}{C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3}$
wo: $\frac{\sqrt{\frac{g_m r_o + 1}{r_o}}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_3}}$
bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{r_o}} (C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3)}{C_1 C_L R_3 r_o \sqrt{g_m + \frac{1}{r_o}}}$
K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L}{C_1 C_L R_3 R_L r_o s^2 + R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s \left(C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{R_3 g_m + \frac{R_3}{r_o} + R_L g_m + \frac{R_L}{r_o}}}{C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_L R_3 R_L g_m r_o + C_L R_3 R_L}$
wo: $\frac{\sqrt{\frac{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L}{r_o}}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L}{r_o}} (C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_L R_3 R_L g_m r_o + C_L R_3 R_L)}{C_1 C_L R_3 R_L r_o \sqrt{R_3 g_m + \frac{R_3}{r_o} + R_L g_m + \frac{R_L}{r_o}}}$

K-LP: $\frac{R_3 R_L}{R_3+R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L g_m r_o + R_L}{C_1 C_3 R_L r_o s^2 + g_m r_o + s\left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_L} r_o \sqrt{g_m + \frac{1}{r_o}}}{C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L}$
wo: $\frac{\sqrt{\frac{g_m r_o + 1}{r_o}}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{r_o}}\left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L\right)}{C_1 C_3 R_L r_o \sqrt{g_m + \frac{1}{r_o}}}$
K-LP: R_L
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.4 LP-4 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L}{g_m r_o + s^2\left(C_1 C_3 R_L r_o + C_1 C_L R_L r_o\right) + s\left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_L} r_o \sqrt{C_3 + C_L}\sqrt{g_m + \frac{1}{r_o}}}{C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L + C_L R_L g_m r_o + C_L R_L}$
wo: $\sqrt{\frac{g_m r_o + 1}{C_1 C_3 R_L r_o + C_1 C_L R_L r_o}}$
bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{C_1 C_3 R_L r_o + C_1 C_L R_L r_o}}\left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L + C_L R_L g_m r_o + C_L R_L\right)}{\sqrt{C_1}\sqrt{R_L} r_o \sqrt{C_3 + C_L}\sqrt{g_m + \frac{1}{r_o}}}$
K-LP: R_L
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.5 LP-5 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L}{C_1 C_3 R_3 R_L r_o s^2 + R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s\left(C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3 R_3 R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{R_3 g_m + \frac{R_3}{r_o} + R_L g_m + \frac{R_L}{r_o}}}{C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3 R_3 R_L}$
wo: $\frac{\sqrt{\frac{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L}{r_o}}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L}{r_o}}\left(C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3 R_3 R_L\right)}{C_1 C_3 R_3 R_L r_o \sqrt{R_3 g_m + \frac{R_3}{r_o} + R_L g_m + \frac{R_L}{r_o}}}$
K-LP: $\frac{R_3 R_L}{R_3+R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.6 LP-6 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3}{g_m r_o + s^2 \left(C_1 C_3 R_3 r_o + C_1 C_L R_3 r_o\right) + s \left(C_1 R_3 + C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3 + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_3} r_o \sqrt{C_3+C_L}\sqrt{g_m+\frac{1}{r_o}}}{C_1 R_3+C_1 r_o+C_3 R_3 g_m r_o+C_3 R_3+C_L R_3 g_m r_o+C_L R_3}$
wo: $\sqrt{\frac{g_m r_o+1}{C_1 C_3 R_3 r_o+C_1 C_L R_3 r_o}}$
bandwidth: $\frac{\sqrt{\frac{g_m r_o+1}{C_1 C_3 R_3 r_o+C_1 C_L R_3 r_o}}\left(C_1 R_3+C_1 r_o+C_3 R_3 g_m r_o+C_3 R_3+C_L R_3 g_m r_o+C_L R_3\right)}{\sqrt{C_1}\sqrt{R_3} r_o \sqrt{C_3+C_L}\sqrt{g_m+\frac{1}{r_o}}}$
K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.7 LP-7 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s^2 \left(C_1 C_3 R_3 R_L r_o + C_1 C_L R_3 R_L r_o\right) + s \left(C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3 R_3 R_L + C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{C_3+C_L}\sqrt{R_3 g_m+\frac{R_3}{r_o}+R_L g_m+\frac{R_L}{r_o}}}{C_1 R_3 R_L+C_1 R_3 r_o+C_1 R_L r_o+C_3 R_3 R_L g_m r_o+C_3 R_3 R_L+C_L R_3 R_L g_m r_o+C_L R_3 R_L}$
wo: $\sqrt{\frac{R_3 g_m r_o+R_3+R_L g_m r_o+R_L}{C_1 C_3 R_3 R_L r_o+C_1 C_L R_3 R_L r_o}}$
bandwidth: $\frac{\sqrt{\frac{R_3 g_m r_o+R_3+R_L g_m r_o+R_L}{C_1 C_3 R_3 R_L r_o+C_1 C_L R_3 R_L r_o}}\left(C_1 R_3 R_L+C_1 R_3 r_o+C_1 R_L r_o+C_3 R_3 R_L g_m r_o+C_3 R_3 R_L+C_L R_3 R_L g_m r_o+C_L R_3 R_L\right)}{\sqrt{C_1}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{C_3+C_L}\sqrt{R_3 g_m+\frac{R_3}{r_o}+R_L g_m+\frac{R_L}{r_o}}}$
K-LP: $\frac{R_3 R_L}{R_3+R_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.8 LP-8 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_3 g_m r_o + R_1 R_3}{C_1 C_L R_1 R_3 r_o s^2 + R_1 g_m r_o + R_1 + R_3 + r_o + s \left(C_1 R_1 R_3 + C_1 R_1 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_3 r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_3} r_o \sqrt{R_1 g_m+\frac{R_1}{r_o}+\frac{R_3}{r_o}+1}}{C_1 R_1 R_3+C_1 R_1 r_o+C_L R_1 R_3 g_m r_o+C_L R_1 R_3+C_L R_3 r_o}$
wo: $\frac{\sqrt{\frac{R_1 g_m r_o+R_1+R_3+r_o}{r_o}}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_3}}$
bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o+R_1+R_3+r_o}{r_o}}\left(C_1 R_1 R_3+C_1 R_1 r_o+C_L R_1 R_3 g_m r_o+C_L R_1 R_3+C_L R_3 r_o\right)}{C_1 C_L R_1 R_3 r_o \sqrt{R_1 g_m+\frac{R_1}{r_o}+\frac{R_3}{r_o}+1}}$
K-LP: $\frac{R_1 R_3 g_m r_o+R_1 R_3}{R_1 g_m r_o+R_1+R_3+r_o}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.9 LP-9 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{R_1R_3R_Lg_mr_o + R_1R_3R_L}{C_1C_LR_1R_3R_Lr_os^2 + R_1R_3g_mr_o + R_1R_3 + R_1R_Lg_mr_o + R_1R_L + R_3R_L + R_3r_o + R_Lr_o + s\left(C_1R_1R_3R_L + C_1R_1R_3r_o + C_1R_1R_Lr_o + C_LR_1R_3R_Lg_mr_o + C_LR_1R_3R_L + C_LR_3R_Lr_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_3}\sqrt{R_L}r_o\sqrt{R_1R_3g_m+\frac{R_1R_3}{r_o}+R_1R_Lg_m+\frac{R_1R_L}{r_o}+\frac{R_3R_L}{r_o}+R_3+R_L}}{C_1R_1R_3R_L+C_1R_1R_3r_o+C_1R_1R_Lr_o+C_LR_1R_3R_Lg_mr_o+C_LR_1R_3R_L+C_LR_3R_Lr_o}$
wo: $\frac{\sqrt{\frac{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}{r_o}}}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_3}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}{r_o}}\left(C_1R_1R_3R_L+C_1R_1R_3r_o+C_1R_1R_Lr_o+C_LR_1R_3R_Lg_mr_o+C_LR_1R_3R_L+C_LR_3R_Lr_o\right)}{C_1C_LR_1R_3R_Lr_o\sqrt{R_1R_3g_m+\frac{R_1R_3}{r_o}+R_1R_Lg_m+\frac{R_1R_L}{r_o}+\frac{R_3R_L}{r_o}+R_3+R_L}}$
K-LP: $\frac{R_1R_3R_Lg_mr_o+R_1R_3R_L}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.10 LP-10 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{1}{C_3s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1R_Lg_mr_o + R_1R_L}{C_1C_3R_1R_Lr_os^2 + R_1g_mr_o + R_1 + R_L + r_o + s\left(C_1R_1R_L + C_1R_1r_o + C_3R_1R_Lg_mr_o + C_3R_1R_L + C_3R_Lr_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_L}r_o\sqrt{R_1g_m+\frac{R_1}{r_o}+\frac{R_L}{r_o}+1}}{C_1R_1R_L+C_1R_1r_o+C_3R_1R_Lg_mr_o+C_3R_1R_L+C_3R_Lr_o}$
wo: $\frac{\sqrt{\frac{R_1g_mr_o+R_1+R_L+r_o}{r_o}}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_1g_mr_o+R_1+R_L+r_o}{r_o}}\left(C_1R_1R_L+C_1R_1r_o+C_3R_1R_Lg_mr_o+C_3R_1R_L+C_3R_Lr_o\right)}{C_1C_3R_1R_Lr_o\sqrt{R_1g_m+\frac{R_1}{r_o}+\frac{R_L}{r_o}+1}}$
K-LP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.11 LP-11 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{1}{C_3s},\ \infty,\ \infty,\ \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_1g_mr_o + R_1}{s^2\left(C_1C_3R_1r_o + C_1C_LR_1r_o\right) + s\left(C_1R_1 + C_3R_1g_mr_o + C_3R_1 + C_3r_o + C_LR_1g_mr_o + C_LR_1 + C_Lr_o\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}r_o\sqrt{C_3+C_L}\sqrt{\frac{1}{r_o}}}{C_1R_1+C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}$
wo: $\sqrt{\frac{1}{C_1C_3R_1r_o+C_1C_LR_1r_o}}$
bandwidth: $\frac{\left(C_1R_1+C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o\right)\sqrt{\frac{1}{C_1C_3R_1r_o+C_1C_LR_1r_o}}}{\sqrt{C_1}\sqrt{R_1}r_o\sqrt{C_3+C_L}\sqrt{\frac{1}{r_o}}}$
K-LP: $R_1g_mr_o + R_1$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.12 LP-12 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{1}{C_3s},\ \infty,\ \infty,\ \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{R_1R_Lg_mr_o + R_1R_L}{R_1g_mr_o + R_1 + R_L + r_o + s^2\left(C_1C_3R_1R_Lr_o + C_1C_LR_1R_Lr_o\right) + s\left(C_1R_1R_L + C_1R_1r_o + C_3R_1R_Lg_mr_o + C_3R_1R_L + C_3R_Lr_o + C_LR_1R_Lg_mr_o + C_LR_1R_L + C_LR_Lr_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_L}r_o\sqrt{C_3+C_L}\sqrt{R_1g_m+\frac{R_1}{r_o}+\frac{R_L}{r_o}+1}}{C_1R_1R_L+C_1R_1r_o+C_3R_1R_Lg_mr_o+C_3R_1R_L+C_3R_Lr_o+C_LR_1R_Lg_mr_o+C_LR_1R_L+C_LR_Lr_o}$
wo: $\sqrt{\frac{R_1g_mr_o+R_1+R_L+r_o}{C_1C_3R_1R_Lr_o+C_1C_LR_1R_Lr_o}}$
bandwidth: $\frac{\sqrt{\frac{R_1g_mr_o+R_1+R_L+r_o}{C_1C_3R_1R_Lr_o+C_1C_LR_1R_Lr_o}}\left(C_1R_1R_L+C_1R_1r_o+C_3R_1R_Lg_mr_o+C_3R_1R_L+C_3R_Lr_o+C_LR_1R_Lg_mr_o+C_LR_1R_L+C_LR_Lr_o\right)}{\sqrt{C_1}\sqrt{R_1}\sqrt{R_L}r_o\sqrt{C_3+C_L}\sqrt{R_1g_m+\frac{R_1}{r_o}+\frac{R_L}{r_o}+1}}$
K-LP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.13 LP-13 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ \infty,\ R_L \right)$

$$H(s) = \frac{R_1R_3R_Lg_mr_o + R_1R_3R_L}{C_1C_3R_1R_3R_Lr_os^2 + R_1R_3g_mr_o + R_1R_3 + R_1R_Lg_mr_o + R_1R_L + R_3R_L + R_3r_o + R_Lr_o + s\left(C_1R_1R_3R_L + C_1R_1R_3r_o + C_1R_1R_Lr_o + C_3R_1R_3R_Lg_mr_o + C_3R_1R_3R_L + C_3R_3R_Lr_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}\sqrt{R_L}r_o\sqrt{R_1R_3g_m+\frac{R_1R_3}{r_o}+R_1R_Lg_m+\frac{R_1R_L}{r_o}+\frac{R_3R_L}{r_o}+R_3+R_L}}{C_1R_1R_3R_L+C_1R_1R_3r_o+C_1R_1R_Lr_o+C_3R_1R_3R_Lg_mr_o+C_3R_1R_3R_L+C_3R_3R_Lr_o}$
wo: $\frac{\sqrt{\frac{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}{r_o}}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}{r_o}}\left(C_1R_1R_3R_L+C_1R_1R_3r_o+C_1R_1R_Lr_o+C_3R_1R_3R_Lg_mr_o+C_3R_1R_3R_L+C_3R_3R_Lr_o\right)}{C_1C_3R_1R_3R_Lr_o\sqrt{R_1R_3g_m+\frac{R_1R_3}{r_o}+R_1R_Lg_m+\frac{R_1R_L}{r_o}+\frac{R_3R_L}{r_o}+R_3+R_L}}$
K-LP: $\frac{R_1R_3R_Lg_mr_o+R_1R_3R_L}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.14 LP-14 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ \infty,\ \frac{1}{C_Ls} \right)$

$$H(s) = \frac{R_1R_3g_mr_o + R_1R_3}{R_1g_mr_o + R_1 + R_3 + r_o + s^2\left(C_1C_3R_1R_3r_o + C_1C_LR_1R_3r_o\right) + s\left(C_1R_1R_3 + C_1R_1r_o + C_3R_1R_3g_mr_o + C_3R_1R_3 + C_3R_3r_o + C_LR_1R_3g_mr_o + C_LR_1R_3 + C_LR_3r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_3}r_o\sqrt{C_3+C_L}\sqrt{R_1g_m+\frac{R_1}{r_o}+\frac{R_3}{r_o}+1}}{C_1R_1R_3+C_1R_1r_o+C_3R_1R_3g_mr_o+C_3R_1R_3+C_3R_3r_o+C_LR_1R_3g_mr_o+C_LR_1R_3+C_LR_3r_o}$
wo: $\sqrt{\frac{R_1g_mr_o+R_1+R_3+r_o}{C_1C_3R_1R_3r_o+C_1C_LR_1R_3r_o}}$
bandwidth: $\frac{\sqrt{\frac{R_1g_mr_o+R_1+R_3+r_o}{C_1C_3R_1R_3r_o+C_1C_LR_1R_3r_o}}\left(C_1R_1R_3+C_1R_1r_o+C_3R_1R_3g_mr_o+C_3R_1R_3+C_3R_3r_o+C_LR_1R_3g_mr_o+C_LR_1R_3+C_LR_3r_o\right)}{\sqrt{C_1}\sqrt{R_1}\sqrt{R_3}r_o\sqrt{C_3+C_L}\sqrt{R_1g_m+\frac{R_1}{r_o}+\frac{R_3}{r_o}+1}}$
K-LP: $\frac{R_1R_3g_mr_o+R_1R_3}{R_1g_mr_o+R_1+R_3+r_o}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.15 LP-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_3 R_L g_m r_o + R_1 R_3 R_L}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o + s^2 \left(C_1 C_3 R_1 R_3 R_L r_o + C_1 C_L R_1 R_3 R_L r_o\right) + s \left(C_1 R_1 R_3 R_L + C_1 R_1 R_3 r_o + C_1 R_1 R_L r_o + C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L + C_3 R_3 R_L r_o + C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L + C_L R_3 R_L r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{R_1}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{C_3+C_L}\sqrt{R_1 R_3 g_m + \frac{R_1 R_3}{r_o} + R_1 R_L g_m + \frac{R_1 R_L}{r_o} + \frac{R_3 R_L}{r_o} + R_3 + R_L}}{C_1 R_1 R_3 R_L + C_1 R_1 R_3 r_o + C_1 R_1 R_L r_o + C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L + C_3 R_3 R_L r_o + C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L + C_L R_3 R_L r_o}$
wo: $\sqrt{\frac{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}{C_1 C_3 R_1 R_3 R_L r_o + C_1 C_L R_1 R_3 R_L r_o}}$
bandwidth: $\frac{\sqrt{\frac{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}{C_1 C_3 R_1 R_3 R_L r_o + C_1 C_L R_1 R_3 R_L r_o}}\left(C_1 R_1 R_3 R_L + C_1 R_1 R_3 r_o + C_1 R_1 R_L r_o + C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L + C_3 R_3 R_L r_o + C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L + C_L R_3 R_L r_o\right)}{\sqrt{C_1}\sqrt{R_1}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{C_3+C_L}\sqrt{R_1 R_3 g_m + \frac{R_1 R_3}{r_o} + R_1 R_L g_m + \frac{R_1 R_L}{r_o} + \frac{R_3 R_L}{r_o} + R_3 + R_L}}$
K-LP: $\frac{R_1 R_3 R_L g_m r_o + R_1 R_3 R_L}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s \left(C_L R_1 R_L g_m r_o + C_L R_1 R_L\right)}{s^2 \left(C_3 C_L R_1 R_L g_m r_o + C_3 C_L R_1 R_L + C_3 C_L R_L r_o\right) + s \left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_L} g_m r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_L}\sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L}\sqrt{R_L} r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}}}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o}$
wo: $\sqrt{\frac{1}{C_3 C_L R_1 R_L g_m r_o + C_3 C_L R_1 R_L + C_3 C_L R_L r_o}}$
bandwidth: $\frac{\left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o\right)\sqrt{\frac{1}{C_3 C_L R_1 R_L g_m r_o + C_3 C_L R_1 R_L + C_3 C_L R_L r_o}}}{\sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_L} g_m r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_L}\sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L}\sqrt{R_L} r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}}}$
K-LP: $R_1 g_m r_o + R_1$
K-HP: 0
K-BP: $\frac{C_L R_1 R_L g_m r_o + C_L R_1 R_L}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_3 g_m r_o + R_1 R_3 + s \left(C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L\right)}{R_1 g_m r_o + R_1 + R_3 + r_o + s^2 \left(C_3 C_L R_1 R_3 R_L g_m r_o + C_3 C_L R_1 R_3 R_L + C_3 C_L R_3 R_L r_o\right) + s \left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_3 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_3 R_L + C_L R_3 r_o + C_L R_L r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} g_m r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L}\sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}}}{C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_3 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_3 R_L + C_L R_3 r_o + C_L R_L r_o}$
wo: $\sqrt{\frac{R_1 g_m r_o + R_1 + R_3 + r_o}{C_3 C_L R_1 R_3 R_L g_m r_o + C_3 C_L R_1 R_3 R_L + C_3 C_L R_3 R_L r_o}}$
bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_3 + r_o}{C_3 C_L R_1 R_3 R_L g_m r_o + C_3 C_L R_1 R_3 R_L + C_3 C_L R_3 R_L r_o}}\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_3 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_3 R_L + C_L R_3 r_o + C_L R_L r_o\right)}{\sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} g_m r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L}\sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}}}$
K-LP: $\frac{R_1 R_3 g_m r_o + R_1 R_3}{R_1 g_m r_o + R_1 + R_3 + r_o}$
K-HP: 0
K-BP: $\frac{C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L}{C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_3 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_3 R_L + C_L R_3 r_o + C_L R_L r_o}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3\right)}{s^2\left(C_3 C_L R_1 R_3 g_m r_o + C_3 C_L R_1 R_3 + C_3 C_L R_3 r_o\right) + s\left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3} g_m r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3} \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L}\sqrt{R_3} r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}}}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

wo: $\sqrt{\frac{1}{C_3 C_L R_1 R_3 g_m r_o + C_3 C_L R_1 R_3 + C_3 C_L R_3 r_o}}$

bandwidth: $\frac{\left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right)\sqrt{\frac{1}{C_3 C_L R_1 R_3 g_m r_o + C_3 C_L R_1 R_3 + C_3 C_L R_3 r_o}}}{\sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3} g_m r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3} \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L}\sqrt{R_3} r_o \sqrt{\frac{1}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: 0

K-BP: $\frac{C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

Qz: None

Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s\left(C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L\right)}{R_1 g_m r_o + R_1 + R_L + r_o + s^2\left(C_3 C_L R_1 R_3 R_L g_m r_o + C_3 C_L R_1 R_3 R_L + C_3 C_L R_3 R_L r_o\right) + s\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_3 R_L + C_3 R_3 r_o + C_3 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} g_m r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}}}{C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_3 R_L + C_3 R_3 r_o + C_3 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

wo: $\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_3 C_L R_1 R_3 R_L g_m r_o + C_3 C_L R_1 R_3 R_L + C_3 C_L R_3 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_3 C_L R_1 R_3 R_L g_m r_o + C_3 C_L R_1 R_3 R_L + C_3 C_L R_3 R_L r_o}}\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_3 R_L + C_3 R_3 r_o + C_3 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o\right)}{\sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} g_m r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{\frac{R_1 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_1}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o} + \frac{r_o}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$

K-HP: 0

K-BP: $\frac{C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L}{C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_3 R_L + C_3 R_3 r_o + C_3 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

Qz: None

Wz: None

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s\left(C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}{g_m r_o + s^2\left(C_1 C_L R_3 R_L + C_1 C_L R_3 r_o + C_1 C_L R_L r_o\right) + s\left(C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3 + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L} R_3 R_L \sqrt{\frac{g_m r_o}{R_3 R_L + R_3 r_o + R_L r_o} + \frac{1}{R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_1}\sqrt{C_L} R_3 r_o \sqrt{\frac{g_m r_o}{R_3 R_L + R_3 r_o + R_L r_o} + \frac{1}{R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_1}\sqrt{C_L} R_L r_o \sqrt{\frac{g_m r_o}{R_3 R_L + R_3 r_o + R_L r_o} + \frac{1}{R_3 R_L + R_3 r_o + R_L r_o}}}{C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3 + C_L R_L g_m r_o + C_L R_L}$

wo: $\sqrt{\frac{g_m r_o + 1}{C_1 C_L R_3 R_L + C_1 C_L R_3 r_o + C_1 C_L R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{C_1 C_L R_3 R_L + C_1 C_L R_3 r_o + C_1 C_L R_L r_o}}\left(C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3 + C_L R_L g_m r_o + C_L R_L\right)}{\sqrt{C_1}\sqrt{C_L} R_3 R_L \sqrt{\frac{g_m r_o}{R_3 R_L + R_3 r_o + R_L r_o} + \frac{1}{R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_1}\sqrt{C_L} R_3 r_o \sqrt{\frac{g_m r_o}{R_3 R_L + R_3 r_o + R_L r_o} + \frac{1}{R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_1}\sqrt{C_L} R_L r_o \sqrt{\frac{g_m r_o}{R_3 R_L + R_3 r_o + R_L r_o} + \frac{1}{R_3 R_L + R_3 r_o + R_L r_o}}}$

K-LP: R_3

K-HP: 0

K-BP: $\frac{C_L R_3 R_L g_m r_o + C_L R_3 R_L}{C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3 + C_L R_L g_m r_o + C_L R_L}$

Qz: None

Wz: None

8.6 X-INVALID-NUMER-6 $Z(s)=\left(\frac{1}{C_1 s},\ \infty,\ R_3+\frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s)=\frac{R_L g_m r_o + R_L + s\left(C_3 R_3 R_L g_m r_o + C_3 R_3 R_L\right)}{g_m r_o + s^2\left(C_1 C_3 R_3 R_L + C_1 C_3 R_3 r_o + C_1 C_3 R_L r_o\right) + s\left(C_1 R_L + C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3 + C_3 R_L g_m r_o + C_3 R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}R_3R_L\sqrt{\frac{g_m r_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}R_3r_o\sqrt{\frac{g_m r_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}R_Lr_o\sqrt{\frac{g_m r_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_3R_L+R_3r_o+R_Lr_o}}}{C_1R_L+C_1r_o+C_3R_3g_mr_o+C_3R_3+C_3R_Lg_mr_o+C_3R_L}$

wo: $\sqrt{\frac{g_m r_o+1}{C_1C_3R_3R_L+C_1C_3R_3r_o+C_1C_3R_Lr_o}}$

bandwidth: $\frac{\sqrt{\frac{g_m r_o+1}{C_1C_3R_3R_L+C_1C_3R_3r_o+C_1C_3R_Lr_o}}\left(C_1R_L+C_1r_o+C_3R_3g_mr_o+C_3R_3+C_3R_Lg_mr_o+C_3R_L\right)}{\sqrt{C_1}\sqrt{C_3}R_3R_L\sqrt{\frac{g_m r_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}R_3r_o\sqrt{\frac{g_m r_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}R_Lr_o\sqrt{\frac{g_m r_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_3R_L+R_3r_o+R_Lr_o}}}$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_3R_3R_Lg_mr_o+C_3R_3R_L}{C_1R_L+C_1r_o+C_3R_3g_mr_o+C_3R_3+C_3R_Lg_mr_o+C_3R_L}$

Qz: None

Wz: None

8.7 X-INVALID-NUMER-7 $Z(s)=\left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L+\frac{1}{C_L s}\right)$

$$H(s)=\frac{R_1 R_3 g_m r_o + R_1 R_3 + s\left(C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L\right)}{R_1 g_m r_o + R_1 + R_3 + r_o + s^2\left(C_1 C_L R_1 R_3 R_L + C_1 C_L R_1 R_3 r_o + C_1 C_L R_1 R_L r_o\right) + s\left(C_1 R_1 R_3 + C_1 R_1 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_3 R_L + C_L R_3 r_o + C_L R_L r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}R_3R_L\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_3}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}R_3r_o\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_3}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}R_Lr_o\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_3}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}}{C_1R_1R_3+C_1R_1r_o+C_LR_1R_3g_mr_o+C_LR_1R_3+C_LR_1R_Lg_mr_o+C_LR_1R_L+C_LR_3R_L+C_LR_3r_o+C_LR_Lr_o}$

wo: $\sqrt{\frac{R_1g_mr_o+R_1+R_3+r_o}{C_1C_LR_1R_3R_L+C_1C_LR_1R_3r_o+C_1C_LR_1R_Lr_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1g_mr_o+R_1+R_3+r_o}{C_1C_LR_1R_3R_L+C_1C_LR_1R_3r_o+C_1C_LR_1R_Lr_o}}\left(C_1R_1R_3+C_1R_1r_o+C_LR_1R_3g_mr_o+C_LR_1R_3+C_LR_1R_Lg_mr_o+C_LR_1R_L+C_LR_3R_L+C_LR_3r_o+C_LR_Lr_o\right)}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}R_3R_L\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_3}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}R_3r_o\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_3}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}R_Lr_o\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_3}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}}$

K-LP: $\frac{R_1R_3g_mr_o+R_1R_3}{R_1g_mr_o+R_1+R_3+r_o}$

K-HP: 0

K-BP: $\frac{C_LR_1R_3R_Lg_mr_o+C_LR_1R_3R_L}{C_1R_1R_3+C_1R_1r_o+C_LR_1R_3g_mr_o+C_LR_1R_3+C_LR_1R_Lg_mr_o+C_LR_1R_L+C_LR_3R_L+C_LR_3r_o+C_LR_Lr_o}$

Qz: None

Wz: None

8.8 X-INVALID-NUMER-8 $Z(s)=\left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ R_3+\frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s)=\frac{R_1 R_L g_m r_o + R_1 R_L + s\left(C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L\right)}{R_1 g_m r_o + R_1 + R_L + r_o + s^2\left(C_1 C_3 R_1 R_3 R_L + C_1 C_3 R_1 R_3 r_o + C_1 C_3 R_1 R_L r_o\right) + s\left(C_1 R_1 R_L + C_1 R_1 r_o + C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_3 R_L + C_3 R_3 r_o + C_3 R_L r_o\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}R_3R_L\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_L}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}R_3r_o\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_L}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}R_Lr_o\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_L}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}}{C_1R_1R_L+C_1R_1r_o+C_3R_1R_3g_mr_o+C_3R_1R_3+C_3R_1R_Lg_mr_o+C_3R_1R_L+C_3R_3R_L+C_3R_3r_o+C_3R_Lr_o}$

wo: $\sqrt{\frac{R_1g_mr_o+R_1+R_L+r_o}{C_1C_3R_1R_3R_L+C_1C_3R_1R_3r_o+C_1C_3R_1R_Lr_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1g_mr_o+R_1+R_L+r_o}{C_1C_3R_1R_3R_L+C_1C_3R_1R_3r_o+C_1C_3R_1R_Lr_o}}\left(C_1R_1R_L+C_1R_1r_o+C_3R_1R_3g_mr_o+C_3R_1R_3+C_3R_1R_Lg_mr_o+C_3R_1R_L+C_3R_3R_L+C_3R_3r_o+C_3R_Lr_o\right)}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}R_3R_L\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_L}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}R_3r_o\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_L}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}R_Lr_o\sqrt{\frac{R_1g_mr_o}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_1}{R_3R_L+R_3r_o+R_Lr_o}+\frac{R_L}{R_3R_L+R_3r_o+R_Lr_o}+\frac{r_o}{R_3R_L+R_3r_o+R_Lr_o}}}$

K-LP: $\frac{R_1R_Lg_mr_o+R_1R_L}{R_1g_mr_o+R_1+R_L+r_o}$

K-HP: 0

K-BP: $\frac{C_3R_1R_3R_Lg_mr_o+C_3R_1R_3R_L}{C_1R_1R_L+C_1R_1r_o+C_3R_1R_3g_mr_o+C_3R_1R_3+C_3R_1R_Lg_mr_o+C_3R_1R_L+C_3R_3R_L+C_3R_3r_o+C_3R_Lr_o}$

Qz: None

Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s\left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3\right)}{g_m r_o + s^2\left(C_1 C_L R_1 R_3 g_m r_o + C_1 C_L R_1 R_3 + C_1 C_L R_3 r_o\right) + s\left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L} R_1 \sqrt{R_3} g_m r_o \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L} R_1 \sqrt{R_3} \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L} \sqrt{R_3} r_o \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}}}{C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3}$

wo: $\sqrt{\frac{g_m r_o + 1}{C_1 C_L R_1 R_3 g_m r_o + C_1 C_L R_1 R_3 + C_1 C_L R_3 r_o}}$

bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{C_1 C_L R_1 R_3 g_m r_o + C_1 C_L R_1 R_3 + C_1 C_L R_3 r_o}}\left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3\right)}{\sqrt{C_1}\sqrt{C_L} R_1 \sqrt{R_3} g_m r_o \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L} R_1 \sqrt{R_3} \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L} \sqrt{R_3} r_o \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: R_3

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3}{C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3}$

Qz: None

Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L + s\left(C_1 R_1 R_3 R_L g_m r_o + C_1 R_1 R_3 R_L\right)}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s^2\left(C_1 C_L R_1 R_3 R_L g_m r_o + C_1 C_L R_1 R_3 R_L + C_1 C_L R_3 R_L r_o\right) + s\left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} g_m r_o \sqrt{\frac{R_3 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} \sqrt{\frac{R_3 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{\frac{R_3 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o}}}{C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_L R_3 R_L g_m r_o + C_L R_3 R_L}$

wo: $\sqrt{\frac{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L}{C_1 C_L R_1 R_3 R_L g_m r_o + C_1 C_L R_1 R_3 R_L + C_1 C_L R_3 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L}{C_1 C_L R_1 R_3 R_L g_m r_o + C_1 C_L R_1 R_3 R_L + C_1 C_L R_3 R_L r_o}}\left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}{\sqrt{C_1}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} g_m r_o \sqrt{\frac{R_3 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L} R_1 \sqrt{R_3}\sqrt{R_L} \sqrt{\frac{R_3 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L} r_o \sqrt{\frac{R_3 g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_3}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{R_L}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: $\frac{R_3 R_L}{R_3 + R_L}$

K-HP: 0

K-BP: $\frac{C_1 R_1 R_3 R_L g_m r_o + C_1 R_1 R_3 R_L}{C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_L R_3 R_L g_m r_o + C_L R_3 R_L}$

Qz: None

Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s\left(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L\right)}{g_m r_o + s^2\left(C_1 C_3 R_1 R_L g_m r_o + C_1 C_3 R_1 R_L + C_1 C_3 R_L r_o\right) + s\left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3} R_1 \sqrt{R_L} g_m r_o \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_3} R_1 \sqrt{R_L} \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_3} \sqrt{R_L} r_o \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}}}{C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L}$

wo: $\sqrt{\frac{g_m r_o + 1}{C_1 C_3 R_1 R_L g_m r_o + C_1 C_3 R_1 R_L + C_1 C_3 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{C_1 C_3 R_1 R_L g_m r_o + C_1 C_3 R_1 R_L + C_1 C_3 R_L r_o}}\left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L\right)}{\sqrt{C_1}\sqrt{C_3} R_1 \sqrt{R_L} g_m r_o \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_3} R_1 \sqrt{R_L} \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}} + \sqrt{C_1}\sqrt{C_3} \sqrt{R_L} r_o \sqrt{\frac{g_m r_o}{R_1 g_m r_o + R_1 + r_o} + \frac{1}{R_1 g_m r_o + R_1 + r_o}}}$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_1 R_1 R_L g_m r_o + C_1 R_1 R_L}{C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L}$

Qz: None

Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s\left(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L\right)}{g_m r_o + s^2\left(C_1 C_3 R_1 R_L g_m r_o + C_1 C_3 R_1 R_L + C_1 C_3 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o\right) + s\left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}C_3R_1\sqrt{R_L}g_mr_o\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{1}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}}+\sqrt{C_1}C_3R_1\sqrt{R_L}\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{1}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}}+\sqrt{C_1}C_3\sqrt{R_L}r_o\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{\cdots}{C_3R_1g_mr_o+\cdots}}\cdots}{\cdots}$

wo: $\sqrt{\frac{g_mr_o+1}{C_1C_3R_1R_Lg_mr_o+C_1C_3R_1R_L+C_1C_3R_Lr_o+C_1C_LR_1R_Lg_mr_o+C_1C_LR_1R_L+C_1C_LR_Lr_o}}$

bandwidth: $\frac{\sqrt{\frac{\cdots}{C_1C_3R_1R_Lg_mr_o+C_1C_3R_1R_L\cdots}}\cdots}{\sqrt{C_1}C_3R_1\sqrt{R_L}g_mr_o\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{1}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}}+\sqrt{C_1}C_3R_1\sqrt{R_L}\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{1}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}}+\sqrt{C_1}C_3\sqrt{R_L}r_o\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\cdots}\cdots}$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_1R_1R_Lg_mr_o+C_1R_1R_L}{C_1R_1g_mr_o+C_1R_1+C_1R_L+C_1r_o+C_3R_Lg_mr_o+C_3R_L+C_LR_Lg_mr_o+C_LR_L}$

Qz: None

Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L + s\left(C_1 R_1 R_3 R_L g_m r_o + C_1 R_1 R_3 R_L\right)}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s^2\left(C_1 C_3 R_1 R_3 R_L g_m r_o + C_1 C_3 R_1 R_3 R_L + C_1 C_3 R_3 R_L r_o\right) + s\left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3 R_3 R_L\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_L}g_mr_o\sqrt{\frac{R_3g_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_3}{R_1g_mr_o+R_1+r_o}+\frac{R_Lg_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_L}{R_1g_mr_o+R_1+r_o}}+\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_L}\sqrt{\frac{R_3g_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_3}{R_1g_mr_o+R_1+r_o}+\frac{R_Lg_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_L}{R_1g_mr_o+R_1+r_o}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_L}r_o\sqrt{\frac{R_3g_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_3}{R_1g_mr_o+R_1+r_o}+\frac{R_Lg_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_L}{R_1g_mr_o+R_1+r_o}}}{C_1R_1R_3g_mr_o+C_1R_1R_3+C_1R_1R_Lg_mr_o+C_1R_1R_L+C_1R_3R_L+C_1R_3r_o+C_1R_Lr_o+C_3R_3R_Lg_mr_o+C_3R_3R_L}$

wo: $\sqrt{\frac{R_3g_mr_o+R_3+R_Lg_mr_o+R_L}{C_1C_3R_1R_3R_Lg_mr_o+C_1C_3R_1R_3R_L+C_1C_3R_3R_Lr_o}}$

bandwidth: $\frac{\sqrt{\frac{R_3g_mr_o+R_3+R_Lg_mr_o+R_L}{C_1C_3R_1R_3R_Lg_mr_o+C_1C_3R_1R_3R_L+C_1C_3R_3R_Lr_o}}\left(C_1R_1R_3g_mr_o+C_1R_1R_3+C_1R_1R_Lg_mr_o+C_1R_1R_L+C_1R_3R_L+C_1R_3r_o+C_1R_Lr_o+C_3R_3R_Lg_mr_o+C_3R_3R_L\right)}{\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_L}g_mr_o\sqrt{\frac{R_3g_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_3}{R_1g_mr_o+R_1+r_o}+\frac{R_Lg_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_L}{R_1g_mr_o+R_1+r_o}}+\sqrt{C_1}\sqrt{C_3}R_1\sqrt{R_3}\sqrt{R_L}\sqrt{\frac{R_3g_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_3}{R_1g_mr_o+R_1+r_o}+\frac{R_Lg_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_L}{R_1g_mr_o+R_1+r_o}}+\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{R_L}r_o\sqrt{\frac{R_3g_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_3}{R_1g_mr_o+R_1+r_o}+\frac{R_Lg_mr_o}{R_1g_mr_o+R_1+r_o}+\frac{R_L}{R_1g_mr_o+R_1+r_o}}}$

K-LP: $\frac{R_3R_L}{R_3+R_L}$

K-HP: 0

K-BP: $\frac{C_1R_1R_3R_Lg_mr_o+C_1R_1R_3R_L}{C_1R_1R_3g_mr_o+C_1R_1R_3+C_1R_1R_Lg_mr_o+C_1R_1R_L+C_1R_3R_L+C_1R_3r_o+C_1R_Lr_o+C_3R_3R_Lg_mr_o+C_3R_3R_L}$

Qz: None

Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s\left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3\right)}{g_m r_o + s^2\left(C_1 C_3 R_1 R_3 g_m r_o + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 r_o + C_1 C_L R_1 R_3 g_m r_o + C_1 C_L R_1 R_3 + C_1 C_L R_3 r_o\right) + s\left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_3 + C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3 + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}C_3R_1\sqrt{R_3}g_mr_o\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{1}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}}+\sqrt{C_1}C_3R_1\sqrt{R_3}\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{1}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}}+\sqrt{C_1}C_3\sqrt{R_3}r_o\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{\cdots}{C_3R_1g_mr_o+C\cdots}}\cdots}{\cdots}$

wo: $\sqrt{\frac{g_mr_o+1}{C_1C_3R_1R_3g_mr_o+C_1C_3R_1R_3+C_1C_3R_3r_o+C_1C_LR_1R_3g_mr_o+C_1C_LR_1R_3+C_1C_LR_3r_o}}$

bandwidth: $\frac{\sqrt{\frac{\cdots}{C_1C_3R_1R_3g_mr_o+C_1C_3R_1R_3\cdots}}\cdots}{\sqrt{C_1}C_3R_1\sqrt{R_3}g_mr_o\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{1}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}}+\sqrt{C_1}C_3R_1\sqrt{R_3}\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\frac{1}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}}+\sqrt{C_1}C_3\sqrt{R_3}r_o\sqrt{\frac{g_mr_o}{C_3R_1g_mr_o+C_3R_1+C_3r_o+C_LR_1g_mr_o+C_LR_1+C_Lr_o}+\cdots}\cdots}$

K-LP: R_3

K-HP: 0

K-BP: $\frac{C_1R_1R_3g_mr_o+C_1R_1R_3}{C_1R_1g_mr_o+C_1R_1+C_1R_3+C_1r_o+C_3R_3g_mr_o+C_3R_3+C_LR_3g_mr_o+C_LR_3}$

Qz: None

Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L + s\left(C_1 R_1 R_3 R_L g_m r_o + C_1 R_1 R_3 R_L\right)}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s^2\left(C_1 C_3 R_1 R_3 R_L g_m r_o + C_1 C_3 R_1 R_3 R_L + C_1 C_3 R_3 R_L r_o + C_1 C_L R_1 R_3 R_L g_m r_o + C_1 C_L R_1 R_3 R_L + C_1 C_L R_3 R_L r_o\right) + s\left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3 R_3 \cdots}$$

Parameters:

Q: $\frac{}{\sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_L} g_m r_o \sqrt{\frac{R_3 g_m r_o}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{R_3}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{R_L g_m r_o}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{R_L}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}} + \sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{R_3 g_m r_o}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{}{C_3 R_1 g_m r_o + C_3 R_1 + C \cdots}}}$

wo: $\sqrt{\frac{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L}{C_1 C_3 R_1 R_3 R_L g_m r_o + C_1 C_3 R_1 R_3 R_L + C_1 C_3 R_3 R_L r_o + C_1 C_L R_1 R_3 R_L g_m r_o + C_1 C_L R_1 R_3 R_L + C_1 C_L R_3 R_L r_o}}$

bandwidth: $\frac{}{\sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_L} g_m r_o \sqrt{\frac{R_3 g_m r_o}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{R_3}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{R_L g_m r_o}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{R_L}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}} + \sqrt{C_1} C_3 R_1 \sqrt{R_3} \sqrt{R_L} \sqrt{\frac{R_3 g_m r_o}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o} + \frac{}{C_3 R_1 g_m \cdots}}}$

K-LP: $\frac{R_3 R_L}{R_3 + R_L}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_3 R_L g_m r_o + C_1 R_1 R_3 R_L}{C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3 R_3 R_L + C_L R_3 R_L g_m r_o + C_L R_3 R_L}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1,\ \infty,\ R_3,\ \infty,\ \infty,\ R_L)$

$$H(s) = \frac{R_1 R_3 R_L g_m r_o + R_1 R_3 R_L}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1,\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_3 g_m r_o + R_1 R_3}{R_1 g_m r_o + R_1 + R_3 + r_o + s\left(C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_3 r_o\right)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1,\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_3 R_L g_m r_o + R_1 R_3 R_L}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o + s\left(C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L + C_L R_3 R_L r_o\right)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1,\ \infty,\ R_3,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_3 g_m r_o + R_1 R_3 + s\left(C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L\right)}{R_1 g_m r_o + R_1 + R_3 + r_o + s\left(C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_3 R_L + C_L R_3 r_o + C_L R_L r_o\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o + s\left(C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_L r_o\right)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1}{s\left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right) + 1}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o + s\left(C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o\right)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2\left(C_L L_L R_1 g_m r_o + C_L L_L R_1\right)}{C_L L_L s^2 + s^3\left(C_3 C_L L_L R_1 g_m r_o + C_3 C_L L_L R_1 + C_3 C_L L_L r_o\right) + s\left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right) + 1}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1 R_3 R_L g_m r_o + R_1 R_3 R_L}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o + s\left(C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L + C_3 R_3 R_L r_o\right)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_3 g_m r_o + R_1 R_3}{R_1 g_m r_o + R_1 + R_3 + r_o + s\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_3 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_3 r_o\right)}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_3 R_L g_m r_o + R_1 R_3 R_L}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o + s\left(C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L + C_3 R_3 R_L r_o + C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L + C_L R_3 R_L r_o\right)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_3 g_m r_o + R_1 R_3 + s^2\left(C_L L_L R_1 R_3 g_m r_o + C_L L_L R_1 R_3\right)}{R_1 g_m r_o + R_1 + R_3 + r_o + s^3\left(C_3 C_L L_L R_1 R_3 g_m r_o + C_3 C_L L_L R_1 R_3 + C_3 C_L L_L R_3 r_o\right) + s^2\left(C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L R_3 + C_L L_L r_o\right) + s\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_3 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_3 r_o\right)}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(R_1,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s\left(C_3 R_1 R_3 R_L g_m r_o + C_3 R_1 R_3 R_L\right)}{R_1 g_m r_o + R_1 + R_L + r_o + s\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_3 R_L + C_3 R_3 r_o + C_3 R_L r_o\right)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(R_1,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^2\left(C_3 L_L R_1 R_3 g_m r_o + C_3 L_L R_1 R_3\right) + s\left(L_L R_1 g_m r_o + L_L R_1\right)}{R_1 g_m r_o + R_1 + r_o + s^3\left(C_3 C_L L_L R_1 R_3 g_m r_o + C_3 C_L L_L R_1 R_3 + C_3 C_L L_L R_3 r_o\right) + s^2\left(C_3 L_L R_1 g_m r_o + C_3 L_L R_1 + C_3 L_L R_3 + C_3 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o\right) + s\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_3 r_o + L_L\right)}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(R_1,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3\left(C_3 C_L L_L R_1 R_3 g_m r_o + C_3 C_L L_L R_1 R_3\right) + s^2\left(C_L L_L R_1 g_m r_o + C_L L_L R_1\right) + s\left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3\right)}{s^3\left(C_3 C_L L_L R_1 g_m r_o + C_3 C_L L_L R_1 + C_3 C_L L_L R_3 + C_3 C_L L_L r_o\right) + s^2\left(C_3 C_L R_1 R_3 g_m r_o + C_3 C_L R_1 R_3 + C_3 C_L R_3 r_o + C_L L_L\right) + s\left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right) + 1}$$

9.16 X-INVALID-ORDER-16 $Z(s) = \left(R_1,\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^2\left(C_L L_3 R_1 R_L g_m r_o + C_L L_3 R_1 R_L\right) + s\left(L_3 R_1 g_m r_o + L_3 R_1\right)}{R_1 g_m r_o + R_1 + r_o + s^3\left(C_3 C_L L_3 R_1 R_L g_m r_o + C_3 C_L L_3 R_1 R_L + C_3 C_L L_3 R_L r_o\right) + s^2\left(C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 r_o + C_L L_3 R_1 g_m r_o + C_L L_3 R_1 + C_L L_3 R_L + C_L L_3 r_o\right) + s\left(C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o + L_3\right)}$$

9.17 X-INVALID-ORDER-17 $Z(s) = \left(R_1,\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3 \left(C_L L_3 L_L R_1 g_m r_o + C_L L_3 L_L R_1\right) + s \left(L_3 R_1 g_m r_o + L_3 R_1\right)}{C_L L_3 L_L s^3 + L_3 s + R_1 g_m r_o + R_1 + r_o + s^4 \left(C_3 C_L L_3 L_L R_1 g_m r_o + C_3 C_L L_3 L_L R_1 + C_3 C_L L_3 L_L r_o\right) + s^2 \left(C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 r_o + C_L L_3 R_1 g_m r_o + C_L L_3 R_1 + C_L L_3 r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o\right)}$$

9.18 X-INVALID-ORDER-18 $Z(s) = \left(R_1,\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 \left(C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1\right)}{C_3 L_3 s^2 + s^3 \left(C_3 C_L L_3 R_1 g_m r_o + C_3 C_L L_3 R_1 + C_3 C_L L_3 r_o\right) + s \left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right) + 1}$$

9.19 X-INVALID-ORDER-19 $Z(s) = \left(R_1,\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s^2 \left(C_3 L_3 R_1 R_L g_m r_o + C_3 L_3 R_1 R_L\right)}{R_1 g_m r_o + R_1 + R_L + r_o + s^3 \left(C_3 C_L L_3 R_1 R_L g_m r_o + C_3 C_L L_3 R_1 R_L + C_3 C_L L_3 R_L r_o\right) + s^2 \left(C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 R_L + C_3 L_3 r_o\right) + s \left(C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o\right)}$$

9.20 X-INVALID-ORDER-20 $Z(s) = \left(R_1,\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 \left(C_3 C_L L_3 R_1 R_L g_m r_o + C_3 C_L L_3 R_1 R_L\right) + s^2 \left(C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1\right) + s \left(C_L R_1 R_L g_m r_o + C_L R_1 R_L\right)}{s^3 \left(C_3 C_L L_3 R_1 g_m r_o + C_3 C_L L_3 R_1 + C_3 C_L L_3 R_L + C_3 C_L L_3 r_o\right) + s^2 \left(C_3 C_L R_1 R_L g_m r_o + C_3 C_L R_1 R_L + C_3 C_L R_L r_o + C_3 L_3\right) + s \left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o\right) + 1}$$

9.21 X-INVALID-ORDER-21 $Z(s) = \left(R_1,\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^3 \left(C_3 L_3 L_L R_1 g_m r_o + C_3 L_3 L_L R_1\right) + s \left(L_L R_1 g_m r_o + L_L R_1\right)}{C_3 L_3 L_L s^3 + L_L s + R_1 g_m r_o + R_1 + r_o + s^4 \left(C_3 C_L L_3 L_L R_1 g_m r_o + C_3 C_L L_3 L_L R_1 + C_3 C_L L_3 L_L r_o\right) + s^2 \left(C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 r_o + C_3 L_L R_1 g_m r_o + C_3 L_L R_1 + C_3 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o\right)}$$

9.22 X-INVALID-ORDER-22 $Z(s) = \left(R_1,\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^4 \left(C_3 C_L L_3 L_L R_1 g_m r_o + C_3 C_L L_3 L_L R_1\right) + s^2 \left(C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_L L_L R_1 g_m r_o + C_L L_L R_1\right)}{C_3 C_L L_3 L_L s^4 + s^3 \left(C_3 C_L L_3 R_1 g_m r_o + C_3 C_L L_3 R_1 + C_3 C_L L_3 r_o + C_3 C_L L_L R_1 g_m r_o + C_3 C_L L_L R_1 + C_3 C_L L_L r_o\right) + s^2 \left(C_3 L_3 + C_L L_L\right) + s \left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right) + 1}$$

9.23 X-INVALID-ORDER-23 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s \left(C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o\right)}$$

9.24 X-INVALID-ORDER-24 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s \left(L_L R_3 g_m r_o + L_L R_3\right)}{C_1 C_L L_L R_3 r_o s^3 + R_3 g_m r_o + R_3 + s^2 \left(C_1 L_L R_3 + C_1 L_L r_o + C_L L_L R_3 g_m r_o + C_L L_L R_3\right) + s \left(C_1 R_3 r_o + L_L g_m r_o + L_L\right)}$$

9.25 X-INVALID-ORDER-25 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^2 \left(C_L L_L R_3 g_m r_o + C_L L_L R_3\right)}{g_m r_o + s^3 \left(C_1 C_L L_L R_3 + C_1 C_L L_L r_o\right) + s^2 \left(C_1 C_L R_3 r_o + C_L L_L g_m r_o + C_L L_L\right) + s \left(C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

9.26 X-INVALID-ORDER-26 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + 1}{s^2 \left(C_1 C_3 r_o + C_1 C_L r_o\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s\left(C_L R_L g_m r_o + C_L R_L\right) + 1}{C_1 C_3 C_L R_L r_o s^3 + s^2\left(C_1 C_3 r_o + C_1 C_L R_L + C_1 C_L r_o + C_3 C_L R_L g_m r_o + C_3 C_L R_L\right) + s\left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s\left(L_L g_m r_o + L_L\right)}{C_1 r_o s + g_m r_o + s^3\left(C_1 C_3 L_L r_o + C_1 C_L L_L r_o\right) + s^2\left(C_1 L_L + C_3 L_L g_m r_o + C_3 L_L + C_L L_L g_m r_o + C_L L_L\right) + 1}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^2\left(C_L L_L g_m r_o + C_L L_L\right) + 1}{C_1 C_3 C_L L_L r_o s^4 + s^3\left(C_1 C_L L_L + C_3 C_L L_L g_m r_o + C_3 C_L L_L\right) + s^2\left(C_1 C_3 r_o + C_1 C_L r_o\right) + s\left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s\left(C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}{C_1 C_3 C_L R_3 R_L r_o s^3 + g_m r_o + s^2\left(C_1 C_3 R_3 r_o + C_1 C_L R_3 R_L + C_1 C_L R_3 r_o + C_1 C_L R_L r_o + C_3 C_L R_3 R_L g_m r_o + C_3 C_L R_3 R_L\right) + s\left(C_1 R_3 + C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3 + C_L R_3 g_m r_o + C_L R_3 + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s\left(L_L R_3 g_m r_o + L_L R_3\right)}{R_3 g_m r_o + R_3 + s^3\left(C_1 C_3 L_L R_3 r_o + C_1 C_L L_L R_3 r_o\right) + s^2\left(C_1 L_L R_3 + C_1 L_L r_o + C_3 L_L R_3 g_m r_o + C_3 L_L R_3 + C_L L_L R_3 g_m r_o + C_L L_L R_3\right) + s\left(C_1 R_3 r_o + L_L g_m r_o + L_L\right)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^2\left(C_L L_L R_3 g_m r_o + C_L L_L R_3\right)}{C_1 C_3 C_L L_L R_3 r_o s^4 + g_m r_o + s^3\left(C_1 C_L L_L R_3 + C_1 C_L L_L r_o + C_3 C_L L_L R_3 g_m r_o + C_3 C_L L_L R_3\right) + s^2\left(C_1 C_3 R_3 r_o + C_1 C_L R_3 r_o + C_L L_L g_m r_o + C_L L_L\right) + s\left(C_1 R_3 + C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3 + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s\left(C_3 R_3 g_m r_o + C_3 R_3\right) + 1}{C_1 C_3 C_L R_3 r_o s^3 + s^2\left(C_1 C_3 R_3 + C_1 C_3 r_o + C_1 C_L r_o + C_3 C_L R_3 g_m r_o + C_3 C_L R_3\right) + s\left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s\left(C_3 R_3 R_L g_m r_o + C_3 R_3 R_L\right)}{C_1 C_3 C_L R_3 R_L r_o s^3 + g_m r_o + s^2\left(C_1 C_3 R_3 R_L + C_1 C_3 R_3 r_o + C_1 C_3 R_L r_o + C_1 C_L R_L r_o + C_3 C_L R_3 R_L g_m r_o + C_3 C_L R_3 R_L\right) + s\left(C_1 R_L + C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3 + C_3 R_L g_m r_o + C_3 R_L + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^2\left(C_3 C_L R_3 R_L g_m r_o + C_3 C_L R_3 R_L\right) + s\left(C_3 R_3 g_m r_o + C_3 R_3 + C_L R_L g_m r_o + C_L R_L\right) + 1}{s^3\left(C_1 C_3 C_L R_3 R_L + C_1 C_3 C_L R_3 r_o + C_1 C_3 C_L R_L r_o\right) + s^2\left(C_1 C_3 R_3 + C_1 C_3 r_o + C_1 C_L R_L + C_1 C_L r_o + C_3 C_L R_3 g_m r_o + C_3 C_L R_3 + C_3 C_L R_L g_m r_o + C_3 C_L R_L\right) + s\left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.36 X-INVALID-ORDER-36 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s^2\left(C_3 L_L R_3 g_m r_o + C_3 L_L R_3\right) + s\left(L_L g_m r_o + L_L\right)}{C_1 C_3 C_L L_L R_3 r_o s^4 + g_m r_o + s^3\left(C_1 C_3 L_L R_3 + C_1 C_3 L_L r_o + C_1 C_L L_L r_o + C_3 C_L L_L R_3 g_m r_o + C_3 C_L L_L R_3\right) + s^2\left(C_1 C_3 R_3 r_o + C_1 L_L + C_3 L_L g_m r_o + C_3 L_L + C_L L_L g_m r_o + C_L L_L\right) + s\left(C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3\right) + 1}$$

9.37 X-INVALID-ORDER-37 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^3\left(C_3 C_L L_L R_3 g_m r_o + C_3 C_L L_L R_3\right) + s^2\left(C_L L_L g_m r_o + C_L L_L\right) + s\left(C_3 R_3 g_m r_o + C_3 R_3\right) + 1}{s^4\left(C_1 C_3 C_L L_L R_3 + C_1 C_3 C_L L_L r_o\right) + s^3\left(C_1 C_3 C_L R_3 r_o + C_1 C_L L_L + C_3 C_L L_L g_m r_o + C_3 C_L L_L\right) + s^2\left(C_1 C_3 R_3 + C_1 C_3 r_o + C_1 C_L r_o + C_3 C_L R_3 g_m r_o + C_3 C_L R_3\right) + s\left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.38 X-INVALID-ORDER-38 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s\left(L_3 R_L g_m r_o + L_3 R_L\right)}{C_1 C_3 L_3 R_L r_o s^3 + R_L g_m r_o + R_L + s^2\left(C_1 L_3 R_L + C_1 L_3 r_o + C_3 L_3 R_L g_m r_o + C_3 L_3 R_L\right) + s\left(C_1 R_L r_o + L_3 g_m r_o + L_3\right)}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{s\left(L_3 g_m r_o + L_3\right)}{C_1 r_o s + g_m r_o + s^3\left(C_1 C_3 L_3 r_o + C_1 C_L L_3 r_o\right) + s^2\left(C_1 L_3 + C_3 L_3 g_m r_o + C_3 L_3 + C_L L_3 g_m r_o + C_L L_3\right) + 1}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{s\left(L_3 R_L g_m r_o + L_3 R_L\right)}{R_L g_m r_o + R_L + s^3\left(C_1 C_3 L_3 R_L r_o + C_1 C_L L_3 R_L r_o\right) + s^2\left(C_1 L_3 R_L + C_1 L_3 r_o + C_3 L_3 R_L g_m r_o + C_3 L_3 R_L + C_L L_3 R_L g_m r_o + C_L L_3 R_L\right) + s\left(C_1 R_L r_o + L_3 g_m r_o + L_3\right)}$$

9.41 X-INVALID-ORDER-41 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^2\left(C_L L_3 R_L g_m r_o + C_L L_3 R_L\right) + s\left(L_3 g_m r_o + L_3\right)}{C_1 C_3 C_L L_3 R_L r_o s^4 + g_m r_o + s^3\left(C_1 C_3 L_3 r_o + C_1 C_L L_3 R_L + C_1 C_L L_3 r_o + C_3 C_L L_3 R_L g_m r_o + C_3 C_L L_3 R_L\right) + s^2\left(C_1 C_L R_L r_o + C_1 L_3 + C_3 L_3 g_m r_o + C_3 L_3 + C_L L_3 g_m r_o + C_L L_3\right) + s\left(C_1 r_o + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

9.42 X-INVALID-ORDER-42 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s\left(L_3 L_L g_m r_o + L_3 L_L\right)}{L_3 g_m r_o + L_3 + L_L g_m r_o + L_L + s^3\left(C_1 C_3 L_3 L_L r_o + C_1 C_L L_3 L_L r_o\right) + s^2\left(C_1 L_3 L_L + C_3 L_3 L_L g_m r_o + C_3 L_3 L_L + C_L L_3 L_L g_m r_o + C_L L_3 L_L\right) + s\left(C_1 L_3 r_o + C_1 L_L r_o\right)}$$

9.43 X-INVALID-ORDER-43 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3\left(C_L L_3 L_L g_m r_o + C_L L_3 L_L\right) + s\left(L_3 g_m r_o + L_3\right)}{C_1 C_3 C_L L_3 L_L r_o s^5 + C_1 r_o s + g_m r_o + s^4\left(C_1 C_L L_3 L_L + C_3 C_L L_3 L_L g_m r_o + C_3 C_L L_3 L_L\right) + s^3\left(C_1 C_3 L_3 r_o + C_1 C_L L_3 r_o + C_1 C_L L_L r_o\right) + s^2\left(C_1 L_3 + C_3 L_3 g_m r_o + C_3 L_3 + C_L L_3 g_m r_o + C_L L_3 + C_L L_L g_m r_o + C_L L_L\right) + 1}$$

9.44 X-INVALID-ORDER-44 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2\left(C_3 L_3 R_L g_m r_o + C_3 L_3 R_L\right)}{g_m r_o + s^3\left(C_1 C_3 L_3 R_L + C_1 C_3 L_3 r_o\right) + s^2\left(C_1 C_3 R_L r_o + C_3 L_3 g_m r_o + C_3 L_3\right) + s\left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L\right) + 1}$$

9.45 X-INVALID-ORDER-45 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^2\left(C_3 L_3 g_m r_o + C_3 L_3\right) + 1}{C_1 C_3 C_L L_3 r_o s^4 + s^3\left(C_1 C_3 L_3 + C_3 C_L L_3 g_m r_o + C_3 C_L L_3\right) + s^2\left(C_1 C_3 r_o + C_1 C_L r_o\right) + s\left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.46 X-INVALID-ORDER-46 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2\left(C_3 L_3 R_L g_m r_o + C_3 L_3 R_L\right)}{C_1 C_3 C_L L_3 R_L r_o s^4 + g_m r_o + s^3\left(C_1 C_3 L_3 R_L + C_1 C_3 L_3 r_o + C_3 C_L L_3 R_L g_m r_o + C_3 C_L L_3 R_L\right) + s^2\left(C_1 C_3 R_L r_o + C_1 C_L R_L r_o + C_3 L_3 g_m r_o + C_3 L_3\right) + s\left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

9.47 X-INVALID-ORDER-47 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^3 \left(C_3 C_L L_3 R_L g_m r_o + C_3 C_L L_3 R_L\right) + s^2 \left(C_3 L_3 g_m r_o + C_3 L_3\right) + s \left(C_L R_L g_m r_o + C_L R_L\right) + 1}{s^4 \left(C_1 C_3 C_L L_3 R_L + C_1 C_3 C_L L_3 r_o\right) + s^3 \left(C_1 C_3 C_L R_L r_o + C_1 C_3 L_3 + C_3 C_L L_3 g_m r_o + C_3 C_L L_3\right) + s^2 \left(C_1 C_3 r_o + C_1 C_L R_L + C_1 C_L r_o + C_3 C_L R_L g_m r_o + C_3 C_L R_L\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s^3 \left(C_3 L_3 L_L g_m r_o + C_3 L_3 L_L\right) + s \left(L_L g_m r_o + L_L\right)}{C_1 C_3 C_L L_3 L_L r_o s^5 + C_1 r_o s + g_m r_o + s^4 \left(C_1 C_3 L_3 L_L + C_3 C_L L_3 L_L g_m r_o + C_3 C_L L_3 L_L\right) + s^3 \left(C_1 C_3 L_3 r_o + C_1 C_3 L_L r_o + C_1 C_L L_L r_o\right) + s^2 \left(C_1 L_L + C_3 L_3 g_m r_o + C_3 L_3 + C_3 L_L g_m r_o + C_3 L_L + C_L L_L g_m r_o + C_L L_L\right) + 1}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^4 \left(C_3 C_L L_3 L_L g_m r_o + C_3 C_L L_3 L_L\right) + s^2 \left(C_3 L_3 g_m r_o + C_3 L_3 + C_L L_L g_m r_o + C_L L_L\right) + 1}{C_1 C_3 C_L L_3 L_L s^5 + s^4 \left(C_1 C_3 C_L L_3 r_o + C_1 C_3 C_L L_L r_o\right) + s^3 \left(C_1 C_3 L_3 + C_1 C_L L_L + C_3 C_L L_3 g_m r_o + C_3 C_L L_3 + C_3 C_L L_L g_m r_o + C_3 C_L L_L\right) + s^2 \left(C_1 C_3 r_o + C_1 C_L r_o\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1 R_3 R_L g_m r_o + R_1 R_3 R_L}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o + s \left(C_1 R_1 R_3 R_L + C_1 R_1 R_3 r_o + C_1 R_1 R_L r_o\right)}$$

9.51 X-INVALID-ORDER-51 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s \left(L_L R_1 R_3 g_m r_o + L_L R_1 R_3\right)}{C_1 C_L L_L R_1 R_3 r_o s^3 + R_1 R_3 g_m r_o + R_1 R_3 + R_3 r_o + s^2 \left(C_1 L_L R_1 R_3 + C_1 L_L R_1 r_o + C_L L_L R_1 R_3 g_m r_o + C_L L_L R_1 R_3 + C_L L_L R_3 r_o\right) + s \left(C_1 R_1 R_3 r_o + L_L R_1 g_m r_o + L_L R_1 + L_L R_3 + L_L r_o\right)}$$

9.52 X-INVALID-ORDER-52 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_3 g_m r_o + R_1 R_3 + s^2 \left(C_L L_L R_1 R_3 g_m r_o + C_L L_L R_1 R_3\right)}{R_1 g_m r_o + R_1 + R_3 + r_o + s^3 \left(C_1 C_L L_L R_1 R_3 + C_1 C_L L_L R_1 r_o\right) + s^2 \left(C_1 C_L R_1 R_3 r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L R_3 + C_L L_L r_o\right) + s \left(C_1 R_1 R_3 + C_1 R_1 r_o + C_L R_1 R_3 g_m r_o + C_L R_1 R_3 + C_L R_3 r_o\right)}$$

9.53 X-INVALID-ORDER-53 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s \left(C_L R_1 R_L g_m r_o + C_L R_1 R_L\right)}{C_1 C_3 C_L R_1 R_L r_o s^3 + s^2 \left(C_1 C_3 R_1 r_o + C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + C_3 C_L R_1 R_L g_m r_o + C_3 C_L R_1 R_L + C_3 C_L R_L r_o\right) + s \left(C_1 R_1 + C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o\right) + 1}$$

9.54 X-INVALID-ORDER-54 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s \left(L_L R_1 g_m r_o + L_L R_1\right)}{R_1 g_m r_o + R_1 + r_o + s^3 \left(C_1 C_3 L_L R_1 r_o + C_1 C_L L_L R_1 r_o\right) + s^2 \left(C_1 L_L R_1 + C_3 L_L R_1 g_m r_o + C_3 L_L R_1 + C_3 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o\right) + s \left(C_1 R_1 r_o + L_L\right)}$$

9.55 X-INVALID-ORDER-55 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 \left(C_L L_L R_1 g_m r_o + C_L L_L R_1\right)}{C_1 C_3 C_L L_L R_1 r_o s^4 + s^3 \left(C_1 C_L L_L R_1 + C_3 C_L L_L R_1 g_m r_o + C_3 C_L L_L R_1 + C_3 C_L L_L r_o\right) + s^2 \left(C_1 C_3 R_1 r_o + C_1 C_L R_1 r_o + C_L L_L\right) + s \left(C_1 R_1 + C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o\right) + 1}$$

9.56 X-INVALID-ORDER-56 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_3 g_m r_o + R_1 R_3 + s \left(C_L R_1 R_3 R_L g_m r_o + C_L R_1 R_3 R_L\right)}{C_1 C_3 C_L R_1 R_3 R_L r_o s^3 + R_1 g_m r_o + R_1 + R_3 + r_o + s^2 \left(C_1 C_3 R_1 R_3 r_o + C_1 C_L R_1 R_3 R_L + C_1 C_L R_1 R_3 r_o + C_1 C_L R_1 R_L r_o + C_3 C_L R_1 R_3 R_L g_m r_o + C_3 C_L R_1 R_3 R_L + C_3 C_L R_3 R_L r_o\right) + s \left(C_1 R_1 R_3 + C_1 R_1 r_o + C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_3 R_3 r_o + C_L R_1 R_3 g_m r_o + C_L R}$$

9.57 X-INVALID-ORDER-57 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ \infty,\ \frac{L_Ls}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{s\left(L_LR_1R_3g_mr_o + L_LR_1R_3\right)}{R_1R_3g_mr_o + R_1R_3 + R_3r_o + s^3\left(C_1C_3L_LR_1R_3r_o + C_1C_LL_LR_1R_3r_o\right) + s^2\left(C_1L_LR_1R_3 + C_1L_LR_1r_o + C_3L_LR_1R_3g_mr_o + C_3L_LR_1R_3 + C_3L_LR_3r_o + C_LL_LR_1R_3g_mr_o + C_LL_LR_1R_3 + C_LL_LR_3r_o\right) + s\left(C_1R_1R_3r_o + L_LR_1g_mr_o + L_LR_1 + L_LR_3 + L_Lr_o\right)}$$

9.58 X-INVALID-ORDER-58 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ \infty,\ L_Ls + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_1R_3g_mr_o + R_1R_3 + s^2\left(C_LL_LR_1R_3g_mr_o + C_LL_LR_1R_3\right)}{C_1C_3C_LL_LR_1R_3r_os^4 + R_1g_mr_o + R_1 + R_3 + r_o + s^3\left(C_1C_LL_LR_1R_3 + C_1C_LL_LR_1r_o + C_3C_LL_LR_1R_3g_mr_o + C_3C_LL_LR_1R_3 + C_3C_LL_LR_3r_o\right) + s^2\left(C_1C_3R_1R_3r_o + C_1C_LR_1R_3r_o + C_LL_LR_1g_mr_o + C_LL_LR_1 + C_LL_LR_3 + C_LL_Lr_o\right) + s\left(C_1R_1R_3 + C_1R_1r_o + C_3R\right.}$$

9.59 X-INVALID-ORDER-59 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_1g_mr_o + R_1 + s\left(C_3R_1R_3g_mr_o + C_3R_1R_3\right)}{C_1C_3C_LR_1R_3r_os^3 + s^2\left(C_1C_3R_1R_3 + C_1C_3R_1r_o + C_1C_LR_1r_o + C_3C_LR_1R_3g_mr_o + C_3C_LR_1R_3 + C_3C_LR_3r_o\right) + s\left(C_1R_1 + C_3R_1g_mr_o + C_3R_1 + C_3R_3 + C_3r_o + C_LR_1g_mr_o + C_LR_1 + C_Lr_o\right) + 1}$$

9.60 X-INVALID-ORDER-60 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{R_1R_Lg_mr_o + R_1R_L + s\left(C_3R_1R_3R_Lg_mr_o + C_3R_1R_3R_L\right)}{C_1C_3C_LR_1R_3R_Lr_os^3 + R_1g_mr_o + R_1 + R_L + r_o + s^2\left(C_1C_3R_1R_3R_L + C_1C_3R_1R_3r_o + C_1C_3R_1R_Lr_o + C_1C_LR_1R_Lr_o + C_3C_LR_1R_3R_Lg_mr_o + C_3C_LR_1R_3R_L + C_3C_LR_3R_Lr_o\right) + s\left(C_1R_1R_L + C_1R_1r_o + C_3R_1R_3g_mr_o + C_3R_1R_3 + C_3R_1R_Lg_mr_o + C_3R_1R_L + C_3\right.}$$

9.61 X-INVALID-ORDER-61 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ R_L + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_1g_mr_o + R_1 + s^2\left(C_3C_LR_1R_3R_Lg_mr_o + C_3C_LR_1R_3R_L\right) + s\left(C_3R_1R_3g_mr_o + C_3R_1R_3 + C_LR_1R_Lg_mr_o + C_LR_1R_L\right)}{s^3\left(C_1C_3C_LR_1R_3R_L + C_1C_3C_LR_1R_3r_o + C_1C_3C_LR_1R_Lr_o\right) + s^2\left(C_1C_3R_1R_3 + C_1C_3R_1r_o + C_1C_LR_1R_L + C_1C_LR_1r_o + C_3C_LR_1R_3g_mr_o + C_3C_LR_1R_3 + C_3C_LR_1R_Lg_mr_o + C_3C_LR_1R_L + C_3C_LR_3R_L + C_3C_LR_3r_o + C_3C_LR_Lr_o\right) + s\left(C_1R_1 + C_3R_1g_mr_o + C_3R_1\right.}$$

9.62 X-INVALID-ORDER-62 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ \frac{L_Ls}{C_LL_Ls^2+1}\right)$

$$H(s) = \frac{s^2\left(C_3L_LR_1R_3g_mr_o + C_3L_LR_1R_3\right) + s\left(L_LR_1g_mr_o + L_LR_1\right)}{C_1C_3C_LL_LR_1R_3r_os^4 + R_1g_mr_o + R_1 + r_o + s^3\left(C_1C_3L_LR_1R_3 + C_1C_3L_LR_1r_o + C_1C_LL_LR_1r_o + C_3C_LL_LR_1R_3g_mr_o + C_3C_LL_LR_1R_3 + C_3C_LL_LR_3r_o\right) + s^2\left(C_1C_3R_1R_3r_o + C_1L_LR_1 + C_3L_LR_1g_mr_o + C_3L_LR_1 + C_3L_LR_3 + C_3L_Lr_o + C_LL_LR_1g_mr_o + C_LL_LR_1\right.}$$

9.63 X-INVALID-ORDER-63 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ R_3 + \frac{1}{C_3s},\ \infty,\ \infty,\ L_Ls + \frac{1}{C_Ls}\right)$

$$H(s) = \frac{R_1g_mr_o + R_1 + s^3\left(C_3C_LL_LR_1R_3g_mr_o + C_3C_LL_LR_1R_3\right) + s^2\left(C_LL_LR_1g_mr_o + C_LL_LR_1\right) + s\left(C_3R_1R_3g_mr_o + C_3R_1R_3\right)}{s^4\left(C_1C_3C_LL_LR_1R_3 + C_1C_3C_LL_LR_1r_o\right) + s^3\left(C_1C_3C_LR_1R_3r_o + C_1C_LL_LR_1 + C_3C_LL_LR_1g_mr_o + C_3C_LL_LR_1 + C_3C_LL_LR_3 + C_3C_LL_Lr_o\right) + s^2\left(C_1C_3R_1R_3 + C_1C_3R_1r_o + C_1C_LR_1r_o + C_3C_LR_1R_3g_mr_o + C_3C_LR_1R_3 + C_3C_LR_3r_o + C_LL_L\right) + s\left(C_1R_1 + C_3R_1g\right.}$$

9.64 X-INVALID-ORDER-64 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{L_3s}{C_3L_3s^2+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s\left(L_3R_1R_Lg_mr_o + L_3R_1R_L\right)}{C_1C_3L_3R_1R_Lr_os^3 + R_1R_Lg_mr_o + R_1R_L + R_Lr_o + s^2\left(C_1L_3R_1R_L + C_1L_3R_1r_o + C_3L_3R_1R_Lg_mr_o + C_3L_3R_1R_L + C_3L_3R_Lr_o\right) + s\left(C_1R_1R_Lr_o + L_3R_1g_mr_o + L_3R_1 + L_3R_L + L_3r_o\right)}$$

9.65 X-INVALID-ORDER-65 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{L_3s}{C_3L_3s^2+1},\ \infty,\ \infty,\ \frac{1}{C_Ls}\right)$

$$H(s) = \frac{s\left(L_3R_1g_mr_o + L_3R_1\right)}{R_1g_mr_o + R_1 + r_o + s^3\left(C_1C_3L_3R_1r_o + C_1C_LL_3R_1r_o\right) + s^2\left(C_1L_3R_1 + C_3L_3R_1g_mr_o + C_3L_3R_1 + C_3L_3r_o + C_LL_3R_1g_mr_o + C_LL_3R_1 + C_LL_3r_o\right) + s\left(C_1R_1r_o + L_3\right)}$$

9.66 X-INVALID-ORDER-66 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \frac{L_3s}{C_3L_3s^2+1},\ \infty,\ \infty,\ \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s) = \frac{s\left(L_3R_1R_Lg_mr_o + L_3R_1R_L\right)}{R_1R_Lg_mr_o + R_1R_L + R_Lr_o + s^3\left(C_1C_3L_3R_1R_Lr_o + C_1C_LL_3R_1R_Lr_o\right) + s^2\left(C_1L_3R_1R_L + C_1L_3R_1r_o + C_3L_3R_1R_Lg_mr_o + C_3L_3R_1R_L + C_3L_3R_Lr_o + C_LL_3R_1R_Lg_mr_o + C_LL_3R_1R_L + C_LL_3R_Lr_o\right) + s\left(C_1R_1R_Lr_o + L_3R_1g_mr_o + L_3R_1 + L_3R_L + L_3r_o\right)}$$

9.67 X-INVALID-ORDER-67 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^2 (C_L L_3 R_1 R_L g_m r_o + C_L L_3 R_1 R_L) + s (L_3 R_1 g_m r_o + L_3 R_1)}{C_1 C_3 C_L L_3 R_1 R_L r_o s^4 + R_1 g_m r_o + R_1 + r_o + s^3 (C_1 C_3 L_3 R_1 r_o + C_1 C_L L_3 R_1 R_L + C_1 C_L L_3 R_1 r_o + C_3 C_L L_3 R_1 R_L g_m r_o + C_3 C_L L_3 R_1 R_L + C_3 C_L L_3 R_L r_o) + s^2 (C_1 C_L R_1 R_L r_o + C_1 L_3 R_1 + C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 r_o + C_L L_3 R_1 g_m r_o + C_L L_3 R_1 + C_L L_3 R_L +}$$

9.68 X-INVALID-ORDER-68 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s (L_3 L_L R_1 g_m r_o + L_3 L_L R_1)}{L_3 R_1 g_m r_o + L_3 R_1 + L_3 r_o + L_L R_1 g_m r_o + L_L R_1 + L_L r_o + s^3 (C_1 C_3 L_3 L_L R_1 r_o + C_1 C_L L_3 L_L R_1 r_o) + s^2 (C_1 L_3 L_L R_1 + C_3 L_3 L_L R_1 g_m r_o + C_3 L_3 L_L R_1 + C_3 L_3 L_L r_o + C_L L_3 L_L R_1 g_m r_o + C_L L_3 L_L R_1 + C_L L_3 L_L r_o) + s (C_1 L_3 R_1 r_o + C_1 L_L R_1 r_o + L_3 L_L)}$$

9.69 X-INVALID-ORDER-69 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3 (C_L L_3 L_L R_1 g_m r_o + C_L L_3 L_L R_1) + s (L_3 R_1 g_m r_o + L_3 R_1)}{C_1 C_3 C_L L_3 L_L R_1 r_o s^5 + R_1 g_m r_o + R_1 + r_o + s^4 (C_1 C_L L_3 L_L R_1 + C_3 C_L L_3 L_L R_1 g_m r_o + C_3 C_L L_3 L_L R_1 + C_3 C_L L_3 L_L r_o) + s^3 (C_1 C_3 L_3 R_1 r_o + C_1 C_L L_3 R_1 r_o + C_1 C_L L_L R_1 r_o + C_L L_3 L_L) + s^2 (C_1 L_3 R_1 + C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 r_o + C_L L_3 R_1 g_m r_o + C_L L_3 R}$$

9.70 X-INVALID-ORDER-70 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s^2 (C_3 L_3 R_1 R_L g_m r_o + C_3 L_3 R_1 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s^3 (C_1 C_3 L_3 R_1 R_L + C_1 C_3 L_3 R_1 r_o) + s^2 (C_1 C_3 R_1 R_L r_o + C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 R_L + C_3 L_3 r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + C_3 R_1 R_L g_m r_o + C_3 R_1 R_L + C_3 R_L r_o)}$$

9.71 X-INVALID-ORDER-71 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 (C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1)}{C_1 C_3 C_L L_3 R_1 r_o s^4 + s^3 (C_1 C_3 L_3 R_1 + C_3 C_L L_3 R_1 g_m r_o + C_3 C_L L_3 R_1 + C_3 C_L L_3 r_o) + s^2 (C_1 C_3 R_1 r_o + C_1 C_L R_1 r_o + C_3 L_3) + s (C_1 R_1 + C_3 R_1 g_m r_o + C_3 R_1 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

9.72 X-INVALID-ORDER-72 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s^2 (C_3 L_3 R_1 R_L g_m r_o + C_3 L_3 R_1 R_L)}{C_1 C_3 C_L L_3 R_1 R_L r_o s^4 + R_1 g_m r_o + R_1 + R_L + r_o + s^3 (C_1 C_3 L_3 R_1 R_L + C_1 C_3 L_3 R_1 r_o + C_3 C_L L_3 R_1 R_L g_m r_o + C_3 C_L L_3 R_1 R_L + C_3 C_L L_3 R_L r_o) + s^2 (C_1 C_3 R_1 R_L r_o + C_1 C_L R_1 R_L r_o + C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 R_L + C_3 L_3 r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + C_3 R_1 R}$$

9.73 X-INVALID-ORDER-73 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_3 C_L L_3 R_1 R_L g_m r_o + C_3 C_L L_3 R_1 R_L) + s^2 (C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1) + s (C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^4 (C_1 C_3 C_L L_3 R_1 R_L + C_1 C_3 C_L L_3 R_1 r_o) + s^3 (C_1 C_3 C_L R_1 R_L r_o + C_1 C_3 L_3 R_1 + C_3 C_L L_3 R_1 g_m r_o + C_3 C_L L_3 R_1 + C_3 C_L L_3 R_L + C_3 C_L L_3 r_o) + s^2 (C_1 C_3 R_1 r_o + C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + C_3 C_L R_1 R_L g_m r_o + C_3 C_L R_1 R_L + C_3 C_L R_L r_o + C_3 L_3) + s (C_1 R_1 + C_3 R_1 g_m}$$

9.74 X-INVALID-ORDER-74 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^3 (C_3 L_3 L_L R_1 g_m r_o + C_3 L_3 L_L R_1) + s (L_L R_1 g_m r_o + L_L R_1)}{C_1 C_3 C_L L_3 L_L R_1 r_o s^5 + R_1 g_m r_o + R_1 + r_o + s^4 (C_1 C_3 L_3 L_L R_1 + C_3 C_L L_3 L_L R_1 g_m r_o + C_3 C_L L_3 L_L R_1 + C_3 C_L L_3 L_L r_o) + s^3 (C_1 C_3 L_3 R_1 r_o + C_1 C_3 L_L R_1 r_o + C_1 C_L L_L R_1 r_o + C_3 L_3 L_L) + s^2 (C_1 L_L R_1 + C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_3 L_3 r_o + C_3 L_L R_1 g_m r_o + C_3 L_L R}$$

9.75 X-INVALID-ORDER-75 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^4 (C_3 C_L L_3 L_L R_1 g_m r_o + C_3 C_L L_3 L_L R_1) + s^2 (C_3 L_3 R_1 g_m r_o + C_3 L_3 R_1 + C_L L_L R_1 g_m r_o + C_L L_L R_1)}{C_1 C_3 C_L L_3 L_L R_1 s^5 + s^4 (C_1 C_3 C_L L_3 R_1 r_o + C_1 C_3 C_L L_L R_1 r_o + C_3 C_L L_3 L_L) + s^3 (C_1 C_3 L_3 R_1 + C_1 C_L L_L R_1 + C_3 C_L L_3 R_1 g_m r_o + C_3 C_L L_3 R_1 + C_3 C_L L_3 r_o + C_3 C_L L_L R_1 g_m r_o + C_3 C_L L_L R_1 + C_3 C_L L_L r_o) + s^2 (C_1 C_3 R_1 r_o + C_1 C_L R_1 r_o + C_3 L_3 + C_L L_L) + s (C_1 R_1}$$

9.76 X-INVALID-ORDER-76 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L + s (C_1 R_1 R_3 R_L g_m r_o + C_1 R_1 R_3 R_L)}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s (C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o)}$$

9.77 X-INVALID-ORDER-77 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s^2 \left(C_1 L_L R_1 R_3 g_m r_o + C_1 L_L R_1 R_3\right) + s \left(L_L R_3 g_m r_o + L_L R_3\right)}{R_3 g_m r_o + R_3 + s^3 \left(C_1 C_L L_L R_1 R_3 g_m r_o + C_1 C_L L_L R_1 R_3 + C_1 C_L L_L R_3 r_o\right) + s^2 \left(C_1 L_L R_1 g_m r_o + C_1 L_L R_1 + C_1 L_L R_3 + C_1 L_L r_o + C_L L_L R_3 g_m r_o + C_L L_L R_3\right) + s \left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_1 R_3 r_o + L_L g_m r_o + L_L\right)}$$

9.78 X-INVALID-ORDER-78 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^3 \left(C_1 C_L L_L R_1 R_3 g_m r_o + C_1 C_L L_L R_1 R_3\right) + s^2 \left(C_L L_L R_3 g_m r_o + C_L L_L R_3\right) + s \left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3\right)}{g_m r_o + s^3 \left(C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L R_3 + C_1 C_L L_L r_o\right) + s^2 \left(C_1 C_L R_1 R_3 g_m r_o + C_1 C_L R_1 R_3 + C_1 C_L R_3 r_o + C_L L_L g_m r_o + C_L L_L\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

9.79 X-INVALID-ORDER-79 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s \left(C_1 R_1 g_m r_o + C_1 R_1\right) + 1}{s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.80 X-INVALID-ORDER-80 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^2 \left(C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 + C_L R_L g_m r_o + C_L R_L\right) + 1}{s^3 \left(C_1 C_3 C_L R_1 R_L g_m r_o + C_1 C_3 C_L R_1 R_L + C_1 C_3 C_L R_L r_o\right) + s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + C_3 C_L R_L g_m r_o + C_3 C_L R_L\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.81 X-INVALID-ORDER-81 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s^2 \left(C_1 L_L R_1 g_m r_o + C_1 L_L R_1\right) + s \left(L_L g_m r_o + L_L\right)}{g_m r_o + s^3 \left(C_1 C_3 L_L R_1 g_m r_o + C_1 C_3 L_L R_1 + C_1 C_3 L_L r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L r_o\right) + s^2 \left(C_1 L_L + C_3 L_L g_m r_o + C_3 L_L + C_L L_L g_m r_o + C_L L_L\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 r_o\right) + 1}$$

9.82 X-INVALID-ORDER-82 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^3 \left(C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1\right) + s^2 \left(C_L L_L g_m r_o + C_L L_L\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1\right) + 1}{s^4 \left(C_1 C_3 C_L L_L R_1 g_m r_o + C_1 C_3 C_L L_L R_1 + C_1 C_3 C_L L_L r_o\right) + s^3 \left(C_1 C_L L_L + C_3 C_L L_L g_m r_o + C_3 C_L L_L\right) + s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.83 X-INVALID-ORDER-83 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^2 \left(C_1 C_L R_1 R_3 R_L g_m r_o + C_1 C_L R_1 R_3 R_L\right) + s \left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3 + C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}{g_m r_o + s^3 \left(C_1 C_3 C_L R_1 R_3 R_L g_m r_o + C_1 C_3 C_L R_1 R_3 R_L + C_1 C_3 C_L R_3 R_L r_o\right) + s^2 \left(C_1 C_3 R_1 R_3 g_m r_o + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 r_o + C_1 C_L R_1 R_3 g_m r_o + C_1 C_L R_1 R_3 + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_3 R_L + C_1 C_L R_3 r_o + C_1 C_L R_L r_o + C_3 C_L R_3 R_L g_m r_o + C_3 C_L}$$

9.84 X-INVALID-ORDER-84 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s^2 \left(C_1 L_L R_1 R_3 g_m r_o + C_1 L_L R_1 R_3\right) + s \left(L_L R_3 g_m r_o + L_L R_3\right)}{R_3 g_m r_o + R_3 + s^3 \left(C_1 C_3 L_L R_1 R_3 g_m r_o + C_1 C_3 L_L R_1 R_3 + C_1 C_3 L_L R_3 r_o + C_1 C_L L_L R_1 R_3 g_m r_o + C_1 C_L L_L R_1 R_3 + C_1 C_L L_L R_3 r_o\right) + s^2 \left(C_1 L_L R_1 g_m r_o + C_1 L_L R_1 + C_1 L_L R_3 + C_1 L_L r_o + C_3 L_L R_3 g_m r_o + C_3 L_L R_3 + C_L L_L R_3 g_m r_o + C_L L_L R_3\right) + s \left(C_1 R_1 R_3 g_m r_o + C\right.}$$

9.85 X-INVALID-ORDER-85 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^3 \left(C_1 C_L L_L R_1 R_3 g_m r_o + C_1 C_L L_L R_1 R_3\right) + s^2 \left(C_L L_L R_3 g_m r_o + C_L L_L R_3\right) + s \left(C_1 R_1 R_3 g_m r_o + C_1 R_1 R_3\right)}{g_m r_o + s^4 \left(C_1 C_3 C_L L_L R_1 R_3 g_m r_o + C_1 C_3 C_L L_L R_1 R_3 + C_1 C_3 C_L L_L R_3 r_o\right) + s^3 \left(C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L R_3 + C_1 C_L L_L r_o + C_3 C_L L_L R_3 g_m r_o + C_3 C_L L_L R_3\right) + s^2 \left(C_1 C_3 R_1 R_3 g_m r_o + C_1 C_3 R_1 R_3 + C_1 C_3 R_3 r_o + C_1 C_L R_1 R_3 g_m r_o + C_1 C_L R_1 R_3 + C_1\right.}$$

9.86 X-INVALID-ORDER-86 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^2 \left(C_1 C_3 R_1 R_3 g_m r_o + C_1 C_3 R_1 R_3\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 + C_3 R_3 g_m r_o + C_3 R_3\right) + 1}{s^3 \left(C_1 C_3 C_L R_1 R_3 g_m r_o + C_1 C_3 C_L R_1 R_3 + C_1 C_3 C_L R_3 r_o\right) + s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o + C_3 C_L R_3 g_m r_o + C_3 C_L R_3\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.87 X-INVALID-ORDER-87 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 \left(C_1 C_3 R_1 R_3 R_L g_m r_o + C_1 C_3 R_1 R_3 R_L\right) + s \left(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_3 R_3 R_L g_m r_o + C_3 R_3 R_L\right)}{g_m r_o + s^3 \left(C_1 C_3 C_L R_1 R_3 R_L g_m r_o + C_1 C_3 C_L R_1 R_3 R_L + C_1 C_3 C_L R_3 R_L r_o\right) + s^2 \left(C_1 C_3 R_1 R_3 g_m r_o + C_1 C_3 R_1 R_3 + C_1 C_3 R_1 R_L g_m r_o + C_1 C_3 R_1 R_L + C_1 C_3 R_3 R_L + C_1 C_3 R_3 r_o + C_1 C_3 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o + C_3 C_L R_3 R_L g_m r_o + C_3 C_L}$$

9.88 X-INVALID-ORDER-88 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^3 \left(C_1 C_3 C_L R_1 R_3 R_L g_m r_o + C_1 C_3 C_L R_1 R_3 R_L\right) + s^2 \left(C_1 C_3 R_1 R_3 g_m r_o + C_1 C_3 R_1 R_3 + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_3 C_L R_3 R_L g_m r_o + C_3 C_L R_3 R_L\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 + C_3 R_3 g_m r_o + C_3 R_3 + C_L}{s^3 \left(C_1 C_3 C_L R_1 R_3 g_m r_o + C_1 C_3 C_L R_1 R_3 + C_1 C_3 C_L R_1 R_L g_m r_o + C_1 C_3 C_L R_1 R_L + C_1 C_3 C_L R_3 R_L + C_1 C_3 C_L R_3 r_o + C_1 C_3 C_L R_L r_o\right) + s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + C_3 C_L R_3 g_m r_o + C_3 C_L R_3 +}$$

9.89 X-INVALID-ORDER-89 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^3 \left(C_1 C_3 L_L R_1 R_3 g_m r_o + C_1 C_3 L_L R_1 R_3\right) + s^2 \left(C_1 L_L R_1 g_m r_o + C_1 L_L R_1 + C_3 L_L R_3 g_m r_o + C_3 L_L R_3\right) + s \left(L_L g_m r_o + L_L\right)}{g_m r_o + s^4 \left(C_1 C_3 C_L L_L R_1 R_3 g_m r_o + C_1 C_3 C_L L_L R_1 R_3 + C_1 C_3 C_L L_L R_3 r_o\right) + s^3 \left(C_1 C_3 L_L R_1 g_m r_o + C_1 C_3 L_L R_1 + C_1 C_3 L_L R_3 + C_1 C_3 L_L r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L r_o + C_3 C_L L_L R_3 g_m r_o + C_3 C_L L_L R_3\right) + s^2 \left(C_1 C_3 R_1 R_3 g_m r_o + C_1 C_3 R_1 R_3 + C_1}$$

9.90 X-INVALID-ORDER-90 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^4 \left(C_1 C_3 C_L L_L R_1 R_3 g_m r_o + C_1 C_3 C_L L_L R_1 R_3\right) + s^3 \left(C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_3 C_L L_L R_3 g_m r_o + C_3 C_L L_L R_3\right) + s^2 \left(C_1 C_3 R_1 R_3 g_m r_o + C_1 C_3 R_1 R_3 + C_L L_L g_m r_o + C_L L_L\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 +}{s^4 \left(C_1 C_3 C_L L_L R_1 g_m r_o + C_1 C_3 C_L L_L R_1 + C_1 C_3 C_L L_L R_3 + C_1 C_3 C_L L_L r_o\right) + s^3 \left(C_1 C_3 C_L R_1 R_3 g_m r_o + C_1 C_3 C_L R_1 R_3 + C_1 C_3 C_L R_3 r_o + C_1 C_L L_L + C_3 C_L L_L g_m r_o + C_3 C_L L_L\right) + s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 R_3 + C_1 C_3 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L}$$

9.91 X-INVALID-ORDER-91 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s^2 \left(C_1 L_3 R_1 R_L g_m r_o + C_1 L_3 R_1 R_L\right) + s \left(L_3 R_L g_m r_o + L_3 R_L\right)}{R_L g_m r_o + R_L + s^3 \left(C_1 C_3 L_3 R_1 R_L g_m r_o + C_1 C_3 L_3 R_1 R_L + C_1 C_3 L_3 R_L r_o\right) + s^2 \left(C_1 L_3 R_1 g_m r_o + C_1 L_3 R_1 + C_1 L_3 R_L + C_1 L_3 r_o + C_3 L_3 R_L g_m r_o + C_3 L_3 R_L\right) + s \left(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_1 R_L r_o + L_3 g_m r_o + L_3\right)}$$

9.92 X-INVALID-ORDER-92 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^2 \left(C_1 L_3 R_1 g_m r_o + C_1 L_3 R_1\right) + s \left(L_3 g_m r_o + L_3\right)}{g_m r_o + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1 + C_1 C_3 L_3 r_o + C_1 C_L L_3 R_1 g_m r_o + C_1 C_L L_3 R_1 + C_1 C_L L_3 r_o\right) + s^2 \left(C_1 L_3 + C_3 L_3 g_m r_o + C_3 L_3 + C_L L_3 g_m r_o + C_L L_3\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 r_o\right) + 1}$$

9.93 X-INVALID-ORDER-93 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{s^2 \left(C_1 L_3 R_1 R_L g_m r_o + C_1 L_3 R_1 R_L\right) + s \left(L_3 R_L g_m r_o + L_3 R_L\right)}{R_L g_m r_o + R_L + s^3 \left(C_1 C_3 L_3 R_1 R_L g_m r_o + C_1 C_3 L_3 R_1 R_L + C_1 C_3 L_3 R_L r_o + C_1 C_L L_3 R_1 R_L g_m r_o + C_1 C_L L_3 R_1 R_L + C_1 C_L L_3 R_L r_o\right) + s^2 \left(C_1 L_3 R_1 g_m r_o + C_1 L_3 R_1 + C_1 L_3 R_L + C_1 L_3 r_o + C_3 L_3 R_L g_m r_o + C_3 L_3 R_L + C_L L_3 R_L g_m r_o + C_L L_3 R_L\right) + s \left(C_1 R_1 R_L g_m r_o + C}$$

9.94 X-INVALID-ORDER-94 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3 \left(C_1 C_L L_3 R_1 R_L g_m r_o + C_1 C_L L_3 R_1 R_L\right) + s^2 \left(C_1 L_3 R_1 g_m r_o + C_1 L_3 R_1 + C_L L_3 R_L g_m r_o + C_L L_3 R_L\right) + s \left(L_3 g_m r_o + L_3\right)}{g_m r_o + s^4 \left(C_1 C_3 C_L L_3 R_1 R_L g_m r_o + C_1 C_3 C_L L_3 R_1 R_L + C_1 C_3 C_L L_3 R_L r_o\right) + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1 + C_1 C_3 L_3 r_o + C_1 C_L L_3 R_1 g_m r_o + C_1 C_L L_3 R_1 + C_1 C_L L_3 R_L + C_1 C_L L_3 r_o + C_3 C_L L_3 R_L g_m r_o + C_3 C_L L_3 R_L\right) + s^2 \left(C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C}$$

9.95 X-INVALID-ORDER-95 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^2 \left(C_1 L_3 L_L R_1 g_m r_o + C_1 L_3 L_L R_1\right) + s \left(L_3 L_L g_m r_o + L_3 L_L\right)}{L_3 g_m r_o + L_3 + L_L g_m r_o + L_L + s^3 \left(C_1 C_3 L_3 L_L R_1 g_m r_o + C_1 C_3 L_3 L_L R_1 + C_1 C_3 L_3 L_L r_o + C_1 C_L L_3 L_L R_1 g_m r_o + C_1 C_L L_3 L_L R_1 + C_1 C_L L_3 L_L r_o\right) + s^2 \left(C_1 L_3 L_L + C_3 L_3 L_L g_m r_o + C_3 L_3 L_L + C_L L_3 L_L g_m r_o + C_L L_3 L_L\right) + s \left(C_1 L_3 R_1 g_m r_o + C_1 L_3 R_1 + C_1 L_3 r_o + C_1 L}$$

9.96 X-INVALID-ORDER-96 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^4 \left(C_1 C_L L_3 L_L R_1 g_m r_o + C_1 C_L L_3 L_L R_1\right) + s^3 \left(C_L L_3 L_L g_m r_o + C_L L_3 L_L\right) + s^2 \left(C_1 L_3 R_1 g_m r_o + C_1 L_3 R_1\right) + s \left(L_3 g_m r_o + L_3\right)}{g_m r_o + s^5 \left(C_1 C_3 C_L L_3 L_L R_1 g_m r_o + C_1 C_3 C_L L_3 L_L R_1 + C_1 C_3 C_L L_3 L_L r_o\right) + s^4 \left(C_1 C_L L_3 L_L + C_3 C_L L_3 L_L g_m r_o + C_3 C_L L_3 L_L\right) + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1 + C_1 C_3 L_3 r_o + C_1 C_L L_3 R_1 g_m r_o + C_1 C_L L_3 R_1 + C_1 C_L L_3 r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C}$$

9.97 X-INVALID-ORDER-97 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 \left(C_1 C_3 L_3 R_1 R_L g_m r_o + C_1 C_3 L_3 R_1 R_L\right) + s^2 \left(C_3 L_3 R_L g_m r_o + C_3 L_3 R_L\right) + s \left(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L\right)}{g_m r_o + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1 + C_1 C_3 L_3 R_L + C_1 C_3 L_3 r_o\right) + s^2 \left(C_1 C_3 R_1 R_L g_m r_o + C_1 C_3 R_1 R_L + C_1 C_3 R_L r_o + C_3 L_3 g_m r_o + C_3 L_3\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L\right) + 1}$$

9.98 X-INVALID-ORDER-98 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1\right) + s^2 \left(C_3 L_3 g_m r_o + C_3 L_3\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1\right) + 1}{s^4 \left(C_1 C_3 C_L L_3 R_1 g_m r_o + C_1 C_3 C_L L_3 R_1 + C_1 C_3 C_L L_3 r_o\right) + s^3 \left(C_1 C_3 L_3 + C_3 C_L L_3 g_m r_o + C_3 C_L L_3\right) + s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.99 X-INVALID-ORDER-99 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 \left(C_1 C_3 L_3 R_1 R_L g_m r_o + C_1 C_3 L_3 R_1 R_L\right) + s^2 \left(C_3 L_3 R_L g_m r_o + C_3 L_3 R_L\right) + s \left(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L\right)}{g_m r_o + s^4 \left(C_1 C_3 C_L L_3 R_1 R_L g_m r_o + C_1 C_3 C_L L_3 R_1 R_L + C_1 C_3 C_L L_3 R_L r_o\right) + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1 + C_1 C_3 L_3 R_L + C_1 C_3 L_3 r_o + C_3 C_L L_3 R_L g_m r_o + C_3 C_L L_3 R_L\right) + s^2 \left(C_1 C_3 R_1 R_L g_m r_o + C_1 C_3 R_1 R_L + C_1 C_3 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C}$$

9.100 X-INVALID-ORDER-100 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^4 \left(C_1 C_3 C_L L_3 R_1 R_L g_m r_o + C_1 C_3 C_L L_3 R_1 R_L\right) + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1 + C_3 C_L L_3 R_L g_m r_o + C_3 C_L L_3 R_L\right) + s^2 \left(C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_3 L_3 g_m r_o + C_3 L_3\right) + s \left(C_1 R_1 g_m r_o + C_1 R_1 + C}{s^4 \left(C_1 C_3 C_L L_3 R_1 g_m r_o + C_1 C_3 C_L L_3 R_1 + C_1 C_3 C_L L_3 R_L + C_1 C_3 C_L L_3 r_o\right) + s^3 \left(C_1 C_3 C_L R_1 R_L g_m r_o + C_1 C_3 C_L R_1 R_L + C_1 C_3 C_L R_L r_o + C_1 C_3 L_3 + C_3 C_L L_3 g_m r_o + C_3 C_L L_3\right) + s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L}$$

9.101 X-INVALID-ORDER-101 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s^4 \left(C_1 C_3 L_3 L_L R_1 g_m r_o + C_1 C_3 L_3 L_L R_1\right) + s^3 \left(C_3 L_3 L_L g_m r_o + C_3 L_3 L_L\right) + s^2 \left(C_1 L_L R_1 g_m r_o + C_1 L_L R_1\right) + s \left(L_L g_m r_o + L_L\right)}{g_m r_o + s^5 \left(C_1 C_3 C_L L_3 L_L R_1 g_m r_o + C_1 C_3 C_L L_3 L_L R_1 + C_1 C_3 C_L L_3 L_L r_o\right) + s^4 \left(C_1 C_3 L_3 L_L + C_3 C_L L_3 L_L g_m r_o + C_3 C_L L_3 L_L\right) + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1 + C_1 C_3 L_3 r_o + C_1 C_3 L_L R_1 g_m r_o + C_1 C_3 L_L R_1 + C_1 C_3 L_L r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C}$$

9.102 X-INVALID-ORDER-102 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^5 \left(C_1 C_3 C_L L_3 L_L R_1 g_m r_o + C_1 C_3 C_L L_3 L_L R_1\right) + s^4 \left(C_3 C_L L_3 L_L g_m r_o + C_3 C_L L_3 L_L\right) + s^3 \left(C_1 C_3 L_3 R_1 g_m r_o + C_1 C_3 L_3 R_1 + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1\right) + s^2 \left(C_3 L_3 g_m r_o + C_3 L_3 + C_L L_L g_m r_o + C_L L_L\right) +}{C_1 C_3 C_L L_3 L_L s^5 + s^4 \left(C_1 C_3 C_L L_3 R_1 g_m r_o + C_1 C_3 C_L L_3 R_1 + C_1 C_3 C_L L_3 r_o + C_1 C_3 C_L L_L R_1 g_m r_o + C_1 C_3 C_L L_L R_1 + C_1 C_3 C_L L_L r_o\right) + s^3 \left(C_1 C_3 L_3 + C_1 C_L L_L + C_3 C_L L_3 g_m r_o + C_3 C_L L_3 + C_3 C_L L_L g_m r_o + C_3 C_L L_L\right) + s^2 \left(C_1 C_3 R_1 g_m r_o + C_1 C_3 R_1 + C_1 C_3 r_o + C_1 C}$$

9.103 X-INVALID-ORDER-103 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{s \left(L_1 R_3 g_m r_o + L_1 R_3\right)}{C_1 C_L L_1 R_3 r_o s^3 + R_3 + r_o + s^2 \left(C_1 L_1 R_3 + C_1 L_1 r_o + C_L L_1 R_3 g_m r_o + C_L L_1 R_3\right) + s \left(C_L R_3 r_o + L_1 g_m r_o + L_1\right)}$$

9.104 X-INVALID-ORDER-104 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{s \left(L_1 R_3 R_L g_m r_o + L_1 R_3 R_L\right)}{C_1 C_L L_1 R_3 R_L r_o s^3 + R_3 R_L + R_3 r_o + R_L r_o + s^2 \left(C_1 L_1 R_3 R_L + C_1 L_1 R_3 r_o + C_1 L_1 R_L r_o + C_L L_1 R_3 R_L g_m r_o + C_L L_1 R_3 R_L\right) + s \left(C_L R_3 R_L r_o + L_1 R_3 g_m r_o + L_1 R_3 + L_1 R_L g_m r_o + L_1 R_L\right)}$$

9.105 X-INVALID-ORDER-105 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^2 \left(C_L L_1 R_3 R_L g_m r_o + C_L L_1 R_3 R_L\right) + s \left(L_1 R_3 g_m r_o + L_1 R_3\right)}{R_3 + r_o + s^3 \left(C_1 C_L L_1 R_3 R_L + C_1 C_L L_1 R_3 r_o + C_1 C_L L_1 R_L r_o\right) + s^2 \left(C_1 L_1 R_3 + C_1 L_1 r_o + C_L L_1 R_3 g_m r_o + C_L L_1 R_3 + C_L L_1 R_L g_m r_o + C_L L_1 R_L\right) + s \left(C_L R_3 R_L + C_L R_3 r_o + C_L R_L r_o + L_1 g_m r_o + L_1\right)}$$

9.106 X-INVALID-ORDER-106 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s^2 \left(L_1 L_L R_3 g_m r_o + L_1 L_L R_3\right)}{C_1 C_L L_1 L_L R_3 r_o s^4 + R_3 r_o + s^3 \left(C_1 L_1 L_L R_3 + C_1 L_1 L_L r_o + C_L L_1 L_L R_3 g_m r_o + C_L L_1 L_L R_3\right) + s^2 \left(C_1 L_1 R_3 r_o + C_L L_L R_3 r_o + L_1 L_L g_m r_o + L_1 L_L\right) + s \left(L_1 R_3 g_m r_o + L_1 R_3 + L_L R_3 + L_L r_o\right)}$$

9.107 X-INVALID-ORDER-107 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ R_3,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3\left(C_L L_1 L_L R_3 g_m r_o + C_L L_1 L_L R_3\right) + s\left(L_1 R_3 g_m r_o + L_1 R_3\right)}{R_3 + r_o + s^4\left(C_1 C_L L_1 L_L R_3 + C_1 C_L L_1 L_L r_o\right) + s^3\left(C_1 C_L L_1 R_3 r_o + C_L L_1 L_L g_m r_o + C_L L_1 L_L\right) + s^2\left(C_1 L_1 R_3 + C_1 L_1 r_o + C_L L_1 R_3 g_m r_o + C_L L_1 R_3 + C_L L_L R_3 + C_L L_L r_o\right) + s\left(C_L R_3 r_o + L_1 g_m r_o + L_1\right)}$$

9.108 X-INVALID-ORDER-108 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s\left(L_1 R_L g_m r_o + L_1 R_L\right)}{C_1 C_3 L_1 R_L r_o s^3 + R_L + r_o + s^2\left(C_1 L_1 R_L + C_1 L_1 r_o + C_3 L_1 R_L g_m r_o + C_3 L_1 R_L\right) + s\left(C_3 R_L r_o + L_1 g_m r_o + L_1\right)}$$

9.109 X-INVALID-ORDER-109 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{s\left(L_1 g_m r_o + L_1\right)}{s^3\left(C_1 C_3 L_1 r_o + C_1 C_L L_1 r_o\right) + s^2\left(C_1 L_1 + C_3 L_1 g_m r_o + C_3 L_1 + C_L L_1 g_m r_o + C_L L_1\right) + s\left(C_3 r_o + C_L r_o\right) + 1}$$

9.110 X-INVALID-ORDER-110 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{s\left(L_1 R_L g_m r_o + L_1 R_L\right)}{R_L + r_o + s^3\left(C_1 C_3 L_1 R_L r_o + C_1 C_L L_1 R_L r_o\right) + s^2\left(C_1 L_1 R_L + C_1 L_1 r_o + C_3 L_1 R_L g_m r_o + C_3 L_1 R_L + C_L L_1 R_L g_m r_o + C_L L_1 R_L\right) + s\left(C_3 R_L r_o + C_L R_L r_o + L_1 g_m r_o + L_1\right)}$$

9.111 X-INVALID-ORDER-111 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^2\left(C_L L_1 R_L g_m r_o + C_L L_1 R_L\right) + s\left(L_1 g_m r_o + L_1\right)}{C_1 C_3 C_L L_1 R_L r_o s^4 + s^3\left(C_1 C_3 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + C_3 C_L L_1 R_L g_m r_o + C_3 C_L L_1 R_L\right) + s^2\left(C_1 L_1 + C_3 C_L R_L r_o + C_3 L_1 g_m r_o + C_3 L_1 + C_L L_1 g_m r_o + C_L L_1\right) + s\left(C_3 r_o + C_L R_L + C_L r_o\right) + 1}$$

9.112 X-INVALID-ORDER-112 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^2\left(L_1 L_L g_m r_o + L_1 L_L\right)}{r_o + s^4\left(C_1 C_3 L_1 L_L r_o + C_1 C_L L_1 L_L r_o\right) + s^3\left(C_1 L_1 L_L + C_3 L_1 L_L g_m r_o + C_3 L_1 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L\right) + s^2\left(C_1 L_1 r_o + C_3 L_L r_o + C_L L_L r_o\right) + s\left(L_1 g_m r_o + L_1 + L_L\right)}$$

9.113 X-INVALID-ORDER-113 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3\left(C_L L_1 L_L g_m r_o + C_L L_1 L_L\right) + s\left(L_1 g_m r_o + L_1\right)}{C_1 C_3 C_L L_1 L_L r_o s^5 + s^4\left(C_1 C_L L_1 L_L + C_3 C_L L_1 L_L g_m r_o + C_3 C_L L_1 L_L\right) + s^3\left(C_1 C_3 L_1 r_o + C_1 C_L L_1 r_o + C_3 C_L L_L r_o\right) + s^2\left(C_1 L_1 + C_3 L_1 g_m r_o + C_3 L_1 + C_L L_1 g_m r_o + C_L L_1 + C_L L_L\right) + s\left(C_3 r_o + C_L r_o\right) + 1}$$

9.114 X-INVALID-ORDER-114 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s\left(L_1 R_3 R_L g_m r_o + L_1 R_3 R_L\right)}{C_1 C_3 L_1 R_3 R_L r_o s^3 + R_3 R_L + R_3 r_o + R_L r_o + s^2\left(C_1 L_1 R_3 R_L + C_1 L_1 R_3 r_o + C_1 L_1 R_L r_o + C_3 L_1 R_3 R_L g_m r_o + C_3 L_1 R_3 R_L\right) + s\left(C_3 R_3 R_L r_o + L_1 R_3 g_m r_o + L_1 R_3 + L_1 R_L g_m r_o + L_1 R_L\right)}$$

9.115 X-INVALID-ORDER-115 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{s\left(L_1 R_3 g_m r_o + L_1 R_3\right)}{R_3 + r_o + s^3\left(C_1 C_3 L_1 R_3 r_o + C_1 C_L L_1 R_3 r_o\right) + s^2\left(C_1 L_1 R_3 + C_1 L_1 r_o + C_3 L_1 R_3 g_m r_o + C_3 L_1 R_3 + C_L L_1 R_3 g_m r_o + C_L L_1 R_3\right) + s\left(C_3 R_3 r_o + C_L R_3 r_o + L_1 g_m r_o + L_1\right)}$$

9.116 X-INVALID-ORDER-116 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{s\left(L_1 R_3 R_L g_m r_o + L_1 R_3 R_L\right)}{R_3 R_L + R_3 r_o + R_L r_o + s^3\left(C_1 C_3 L_1 R_3 R_L r_o + C_1 C_L L_1 R_3 R_L r_o\right) + s^2\left(C_1 L_1 R_3 R_L + C_1 L_1 R_3 r_o + C_1 L_1 R_L r_o + C_3 L_1 R_3 R_L g_m r_o + C_3 L_1 R_3 R_L + C_L L_1 R_3 R_L g_m r_o + C_L L_1 R_3 R_L\right) + s\left(C_3 R_3 R_L r_o + C_L R_3 R_L r_o + L_1 R_3 g_m r_o + L_1 R_3 + L_1 R_L g_m r_o + L_1 R_L\right)}$$

9.117 X-INVALID-ORDER-117 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^2 \left(C_L L_1 R_3 R_L g_m r_o + C_L L_1 R_3 R_L\right) + s \left(L_1 R_3 g_m r_o + L_1 R_3\right)}{C_1 C_3 C_L L_1 R_3 R_L r_o s^4 + R_3 + r_o + s^3 \left(C_1 C_3 L_1 R_3 r_o + C_1 C_L L_1 R_3 R_L + C_1 C_L L_1 R_3 r_o + C_1 C_L L_1 R_L r_o + C_3 C_L L_1 R_3 R_L g_m r_o + C_3 C_L L_1 R_3 R_L\right) + s^2 \left(C_1 L_1 R_3 + C_1 L_1 r_o + C_3 C_L R_3 R_L r_o + C_3 L_1 R_3 g_m r_o + C_3 L_1 R_3 + C_L L_1 R_3 g_m r_o + C_L L_1 R_3 + C_L L_1 R_L g_m r_o$$

9.118 X-INVALID-ORDER-118 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^2 \left(L_1 L_L R_3 g_m r_o + L_1 L_L R_3\right)}{R_3 r_o + s^4 \left(C_1 C_3 L_1 L_L R_3 r_o + C_1 C_L L_1 L_L R_3 r_o\right) + s^3 \left(C_1 L_1 L_L R_3 + C_1 L_1 L_L r_o + C_3 L_1 L_L R_3 g_m r_o + C_3 L_1 L_L R_3 + C_L L_1 L_L R_3 g_m r_o + C_L L_1 L_L R_3\right) + s^2 \left(C_1 L_1 R_3 r_o + C_3 L_L R_3 r_o + C_L L_L R_3 r_o + L_1 L_L g_m r_o + L_1 L_L\right) + s \left(L_1 R_3 g_m r_o + L_1 R_3 + L_L R_3 + L_L r_o\right)}$$

9.119 X-INVALID-ORDER-119 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3 \left(C_L L_1 L_L R_3 g_m r_o + C_L L_1 L_L R_3\right) + s \left(L_1 R_3 g_m r_o + L_1 R_3\right)}{C_1 C_3 C_L L_1 L_L R_3 r_o s^5 + R_3 + r_o + s^4 \left(C_1 C_L L_1 L_L R_3 + C_1 C_L L_1 L_L r_o + C_3 C_L L_1 L_L R_3 g_m r_o + C_3 C_L L_1 L_L R_3\right) + s^3 \left(C_1 C_3 L_1 R_3 r_o + C_1 C_L L_1 R_3 r_o + C_3 C_L L_L R_3 r_o + C_L L_1 L_L g_m r_o + C_L L_1 L_L\right) + s^2 \left(C_1 L_1 R_3 + C_1 L_1 r_o + C_3 L_1 R_3 g_m r_o + C_3 L_1 R_3 + C_L L_1 R_3 g_m r_o$$

9.120 X-INVALID-ORDER-120 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s^2 \left(C_3 L_1 R_3 R_L g_m r_o + C_3 L_1 R_3 R_L\right) + s \left(L_1 R_L g_m r_o + L_1 R_L\right)}{R_L + r_o + s^3 \left(C_1 C_3 L_1 R_3 R_L + C_1 C_3 L_1 R_3 r_o + C_1 C_3 L_1 R_L r_o\right) + s^2 \left(C_1 L_1 R_L + C_1 L_1 r_o + C_3 L_1 R_3 g_m r_o + C_3 L_1 R_3 + C_3 L_1 R_L g_m r_o + C_3 L_1 R_L\right) + s \left(C_3 R_3 R_L + C_3 R_3 r_o + C_3 R_L r_o + L_1 g_m r_o + L_1\right)}$$

9.121 X-INVALID-ORDER-121 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^2 \left(C_3 L_1 R_3 g_m r_o + C_3 L_1 R_3\right) + s \left(L_1 g_m r_o + L_1\right)}{C_1 C_3 C_L L_1 R_3 r_o s^4 + s^3 \left(C_1 C_3 L_1 R_3 + C_1 C_3 L_1 r_o + C_1 C_L L_1 r_o + C_3 C_L L_1 R_3 g_m r_o + C_3 C_L L_1 R_3\right) + s^2 \left(C_1 L_1 + C_3 C_L R_3 r_o + C_3 L_1 g_m r_o + C_3 L_1 + C_L L_1 g_m r_o + C_L L_1\right) + s \left(C_3 R_3 + C_3 r_o + C_L r_o\right) + 1}$$

9.122 X-INVALID-ORDER-122 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{s^2 \left(C_3 L_1 R_3 R_L g_m r_o + C_3 L_1 R_3 R_L\right) + s \left(L_1 R_L g_m r_o + L_1 R_L\right)}{C_1 C_3 C_L L_1 R_3 R_L r_o s^4 + R_L + r_o + s^3 \left(C_1 C_3 L_1 R_3 R_L + C_1 C_3 L_1 R_3 r_o + C_1 C_3 L_1 R_L r_o + C_1 C_L L_1 R_L r_o + C_3 C_L L_1 R_3 R_L g_m r_o + C_3 C_L L_1 R_3 R_L\right) + s^2 \left(C_1 L_1 R_L + C_1 L_1 r_o + C_3 C_L R_3 R_L r_o + C_3 L_1 R_3 g_m r_o + C_3 L_1 R_3 + C_3 L_1 R_L g_m r_o + C_3 L_1 R_L + C_L L_1 R_L g_m r_o$$

9.123 X-INVALID-ORDER-123 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3 \left(C_3 C_L L_1 R_3 R_L g_m r_o + C_3 C_L L_1 R_3 R_L\right) + s^2 \left(C_3 L_1 R_3 g_m r_o + C_3 L_1 R_3 + C_L L_1 R_L g_m r_o + C_L L_1 R_L\right) + s \left(L_1 g_m r_o + L_1\right)}{s^4 \left(C_1 C_3 C_L L_1 R_3 R_L + C_1 C_3 C_L L_1 R_3 r_o + C_1 C_3 C_L L_1 R_L r_o\right) + s^3 \left(C_1 C_3 L_1 R_3 + C_1 C_3 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + C_3 C_L L_1 R_3 g_m r_o + C_3 C_L L_1 R_3 + C_3 C_L L_1 R_L g_m r_o + C_3 C_L L_1 R_L\right) + s^2 \left(C_1 L_1 + C_3 C_L R_3 R_L + C_3 C_L R_3 r_o + C_3 C_L R_L r_o + C_3 L_1 g_m r_o + C_3 L_1$$

9.124 X-INVALID-ORDER-124 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^3 \left(C_3 L_1 L_L R_3 g_m r_o + C_3 L_1 L_L R_3\right) + s^2 \left(L_1 L_L g_m r_o + L_1 L_L\right)}{C_1 C_3 C_L L_1 L_L R_3 r_o s^5 + r_o + s^4 \left(C_1 C_3 L_1 L_L R_3 + C_1 C_3 L_1 L_L r_o + C_1 C_L L_1 L_L r_o + C_3 C_L L_1 L_L R_3 g_m r_o + C_3 C_L L_1 L_L R_3\right) + s^3 \left(C_1 C_3 L_1 R_3 r_o + C_1 L_1 L_L + C_3 C_L L_L R_3 r_o + C_3 L_1 L_L g_m r_o + C_3 L_1 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L\right) + s^2 \left(C_1 L_1 r_o + C_3 L_1 R_3 g_m r_o$$

9.125 X-INVALID-ORDER-125 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^4 \left(C_3 C_L L_1 L_L R_3 g_m r_o + C_3 C_L L_1 L_L R_3\right) + s^3 \left(C_L L_1 L_L g_m r_o + C_L L_1 L_L\right) + s^2 \left(C_3 L_1 R_3 g_m r_o + C_3 L_1 R_3\right) + s \left(L_1 g_m r_o + L_1\right)}{s^5 \left(C_1 C_3 C_L L_1 L_L R_3 + C_1 C_3 C_L L_1 L_L r_o\right) + s^4 \left(C_1 C_3 C_L L_1 R_3 r_o + C_1 C_L L_1 L_L + C_3 C_L L_1 L_L g_m r_o + C_3 C_L L_1 L_L\right) + s^3 \left(C_1 C_3 L_1 R_3 + C_1 C_3 L_1 r_o + C_1 C_L L_1 r_o + C_3 C_L L_1 R_3 g_m r_o + C_3 C_L L_1 R_3 + C_3 C_L L_L R_3 + C_3 C_L L_L r_o\right) + s^2 \left(C_1 L_1 + C_3 C_L R_3 r_o + C_3 L_1 g_m r_o$$

9.126 X-INVALID-ORDER-126 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s^2 \left(L_1 L_3 R_L g_m r_o + L_1 L_3 R_L\right)}{C_1 C_3 L_1 L_3 R_L r_o s^4 + R_L r_o + s^3 \left(C_1 L_1 L_3 R_L + C_1 L_1 L_3 r_o + C_3 L_1 L_3 R_L g_m r_o + C_3 L_1 L_3 R_L\right) + s^2 \left(C_1 L_1 R_L r_o + C_3 L_3 R_L r_o + L_1 L_3 g_m r_o + L_1 L_3\right) + s \left(L_1 R_L g_m r_o + L_1 R_L + L_3 R_L + L_3 r_o\right)}$$

9.127 X-INVALID-ORDER-127 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^2 \left(L_1 L_3 g_m r_o + L_1 L_3\right)}{r_o + s^4 \left(C_1 C_3 L_1 L_3 r_o + C_1 C_L L_1 L_3 r_o\right) + s^3 \left(C_1 L_1 L_3 + C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3 + C_L L_1 L_3 g_m r_o + C_L L_1 L_3\right) + s^2 \left(C_1 L_1 r_o + C_3 L_3 r_o + C_L L_3 r_o\right) + s \left(L_1 g_m r_o + L_1 + L_3\right)}$$

9.128 X-INVALID-ORDER-128 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{s^2 \left(L_1 L_3 R_L g_m r_o + L_1 L_3 R_L\right)}{R_L r_o + s^4 \left(C_1 C_3 L_1 L_3 R_L r_o + C_1 C_L L_1 L_3 R_L r_o\right) + s^3 \left(C_1 L_1 L_3 R_L + C_1 L_1 L_3 r_o + C_3 L_1 L_3 R_L g_m r_o + C_3 L_1 L_3 R_L + C_L L_1 L_3 R_L g_m r_o + C_L L_1 L_3 R_L\right) + s^2 \left(C_1 L_1 R_L r_o + C_3 L_3 R_L r_o + C_L L_3 R_L r_o + L_1 L_3 g_m r_o + L_1 L_3\right) + s \left(L_1 R_L g_m r_o + L_1 R_L + L_3 R_L + L_3 r_o\right)}$$

9.129 X-INVALID-ORDER-129 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^3 \left(C_L L_1 L_3 R_L g_m r_o + C_L L_1 L_3 R_L\right) + s^2 \left(L_1 L_3 g_m r_o + L_1 L_3\right)}{C_1 C_3 C_L L_1 L_3 R_L r_o s^5 + r_o + s^4 \left(C_1 C_3 L_1 L_3 r_o + C_1 C_L L_1 L_3 R_L + C_1 C_L L_1 L_3 r_o + C_3 C_L L_1 L_3 R_L g_m r_o + C_3 C_L L_1 L_3 R_L\right) + s^3 \left(C_1 C_L L_1 R_L r_o + C_1 L_1 L_3 + C_3 C_L L_3 R_L r_o + C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3 + C_L L_1 L_3 g_m r_o + C_L L_1 L_3\right) + s^2 \left(C_1 L_1 r_o + C_3 L_3 r_o + C_L L_1 R_L g_m r_o\right.}$$

9.130 X-INVALID-ORDER-130 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^2 \left(L_1 L_3 L_L g_m r_o + L_1 L_3 L_L\right)}{L_3 r_o + L_L r_o + s^4 \left(C_1 C_3 L_1 L_3 L_L r_o + C_1 C_L L_1 L_3 L_L r_o\right) + s^3 \left(C_1 L_1 L_3 L_L + C_3 L_1 L_3 L_L g_m r_o + C_3 L_1 L_3 L_L + C_L L_1 L_3 L_L g_m r_o + C_L L_1 L_3 L_L\right) + s^2 \left(C_1 L_1 L_3 r_o + C_1 L_1 L_L r_o + C_3 L_3 L_L r_o + C_L L_3 L_L r_o\right) + s \left(L_1 L_3 g_m r_o + L_1 L_3 + L_1 L_L g_m r_o + L_1 L_L + L_3 L_L\right)}$$

9.131 X-INVALID-ORDER-131 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2 + 1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^4 \left(C_L L_1 L_3 L_L g_m r_o + C_L L_1 L_3 L_L\right) + s^2 \left(L_1 L_3 g_m r_o + L_1 L_3\right)}{C_1 C_3 C_L L_1 L_3 L_L r_o s^6 + r_o + s^5 \left(C_1 C_L L_1 L_3 L_L + C_3 C_L L_1 L_3 L_L g_m r_o + C_3 C_L L_1 L_3 L_L\right) + s^4 \left(C_1 C_3 L_1 L_3 r_o + C_1 C_L L_1 L_3 r_o + C_1 C_L L_1 L_L r_o + C_3 C_L L_3 L_L r_o\right) + s^3 \left(C_1 L_1 L_3 + C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3 + C_L L_1 L_3 g_m r_o + C_L L_1 L_3 + C_L L_1 L_L g_m r_o + C_L L_1 L_L + C_L L_3 L_L\right.}$$

9.132 X-INVALID-ORDER-132 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L \right)$

$$H(s) = \frac{s^3 \left(C_3 L_1 L_3 R_L g_m r_o + C_3 L_1 L_3 R_L\right) + s \left(L_1 R_L g_m r_o + L_1 R_L\right)}{R_L + r_o + s^4 \left(C_1 C_3 L_1 L_3 R_L + C_1 C_3 L_1 L_3 r_o\right) + s^3 \left(C_1 C_3 L_1 R_L r_o + C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3\right) + s^2 \left(C_1 L_1 R_L + C_1 L_1 r_o + C_3 L_1 R_L g_m r_o + C_3 L_1 R_L + C_3 L_3 R_L + C_3 L_3 r_o\right) + s \left(C_3 R_L r_o + L_1 g_m r_o + L_1\right)}$$

9.133 X-INVALID-ORDER-133 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^3 \left(C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3\right) + s \left(L_1 g_m r_o + L_1\right)}{C_1 C_3 C_L L_1 L_3 r_o s^5 + s^4 \left(C_1 C_3 L_1 L_3 + C_3 C_L L_1 L_3 g_m r_o + C_3 C_L L_1 L_3\right) + s^3 \left(C_1 C_3 L_1 r_o + C_1 C_L L_1 r_o + C_3 C_L L_3 r_o\right) + s^2 \left(C_1 L_1 + C_3 L_1 g_m r_o + C_3 L_1 + C_3 L_3 + C_L L_1 g_m r_o + C_L L_1\right) + s \left(C_3 r_o + C_L r_o\right) + 1}$$

9.134 X-INVALID-ORDER-134 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{s^3 \left(C_3 L_1 L_3 R_L g_m r_o + C_3 L_1 L_3 R_L\right) + s \left(L_1 R_L g_m r_o + L_1 R_L\right)}{C_1 C_3 C_L L_1 L_3 R_L r_o s^5 + R_L + r_o + s^4 \left(C_1 C_3 L_1 L_3 R_L + C_1 C_3 L_1 L_3 r_o + C_3 C_L L_1 L_3 R_L g_m r_o + C_3 C_L L_1 L_3 R_L\right) + s^3 \left(C_1 C_3 L_1 R_L r_o + C_1 C_L L_1 R_L r_o + C_3 C_L L_3 R_L r_o + C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3\right) + s^2 \left(C_1 L_1 R_L + C_1 L_1 r_o + C_3 L_1 R_L g_m r_o + C_3 L_1 R_L + C_3 L_3 R_L\right.}$$

9.135 X-INVALID-ORDER-135 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{s^4 \left(C_3 C_L L_1 L_3 R_L g_m r_o + C_3 C_L L_1 L_3 R_L\right) + s^3 \left(C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3\right) + s^2 \left(C_L L_1 R_L g_m r_o + C_L L_1 R_L\right) + s \left(L_1 g_m r_o + L_1\right)}{s^5 \left(C_1 C_3 C_L L_1 L_3 R_L + C_1 C_3 C_L L_1 L_3 r_o\right) + s^4 \left(C_1 C_3 C_L L_1 R_L r_o + C_1 C_3 L_1 L_3 + C_3 C_L L_1 L_3 g_m r_o + C_3 C_L L_1 L_3\right) + s^3 \left(C_1 C_3 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + C_3 C_L L_1 R_L g_m r_o + C_3 C_L L_1 R_L + C_3 C_L L_3 R_L + C_3 C_L L_3 r_o\right) + s^2 \left(C_1 L_1 + C_3 C_L R_L r_o + C_3 L_1 g_m r_o\right.}$$

9.136 X-INVALID-ORDER-136 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^4 \left(C_3 L_1 L_3 L_L g_m r_o + C_3 L_1 L_3 L_L\right) + s^2 \left(L_1 L_L g_m r_o + L_1 L_L\right)}{C_1 C_3 C_L L_1 L_3 L_L r_o s^6 + r_o + s^5 \left(C_1 C_3 L_1 L_3 L_L + C_3 C_L L_1 L_3 L_L g_m r_o + C_3 C_L L_1 L_3 L_L\right) + s^4 \left(C_1 C_3 L_1 L_3 r_o + C_1 C_3 L_1 L_L r_o + C_1 C_L L_1 L_L r_o + C_3 C_L L_3 L_L r_o\right) + s^3 \left(C_1 L_1 L_L + C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3 + C_3 L_1 L_L g_m r_o + C_3 L_1 L_L + C_3 L_3 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L\right.}$$

9.137 X-INVALID-ORDER-137 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^5\left(C_3 C_L L_1 L_3 L_L g_m r_o + C_3 C_L L_1 L_3 L_L\right) + s^3\left(C_3 L_1 L_3 g_m r_o + C_3 L_1 L_3 + C_L L_1 L_L g_m r_o + C_L L_1 L_L\right) + s\left(L_1 g_m r_o + L_1\right)}{C_1 C_3 C_L L_1 L_3 L_L s^6 + s^5\left(C_1 C_3 C_L L_1 L_3 r_o + C_1 C_3 C_L L_1 L_L r_o\right) + s^4\left(C_1 C_3 L_1 L_3 + C_1 C_L L_1 L_L + C_3 C_L L_1 L_3 g_m r_o + C_3 C_L L_1 L_3 + C_3 C_L L_1 L_L g_m r_o + C_3 C_L L_1 L_L + C_3 C_L L_3 L_L\right) + s^3\left(C_1 C_3 L_1 r_o + C_1 C_L L_1 r_o + C_3 C_L L_3 r_o + C_3 C_L L_L r_o\right) + s^2 (C_1 L_1 + C_3 L_1 g_m r_o + C_3}$$

9.138 X-INVALID-ORDER-138 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^2\left(C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3\right)}{g_m r_o + s^3\left(C_1 C_L L_1 R_3 g_m r_o + C_1 C_L L_1 R_3\right) + s^2\left(C_1 C_L R_3 r_o + C_1 L_1 g_m r_o + C_1 L_1\right) + s\left(C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

9.139 X-INVALID-ORDER-139 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L + s^2\left(C_1 L_1 R_3 R_L g_m r_o + C_1 L_1 R_3 R_L\right)}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s^3\left(C_1 C_L L_1 R_3 R_L g_m r_o + C_1 C_L L_1 R_3 R_L\right) + s^2\left(C_1 C_L R_3 R_L r_o + C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3 + C_1 L_1 R_L g_m r_o + C_1 L_1 R_L\right) + s\left(C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}$$

9.140 X-INVALID-ORDER-140 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^3\left(C_1 C_L L_1 R_3 R_L g_m r_o + C_1 C_L L_1 R_3 R_L\right) + s^2\left(C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3\right) + s\left(C_L R_3 R_L g_m r_o + C_L R_3 R_L\right)}{g_m r_o + s^3\left(C_1 C_L L_1 R_3 g_m r_o + C_1 C_L L_1 R_3 + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L\right) + s^2\left(C_1 C_L R_3 R_L + C_1 C_L R_3 r_o + C_1 C_L R_L r_o + C_1 L_1 g_m r_o + C_1 L_1\right) + s\left(C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3 + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

9.141 X-INVALID-ORDER-141 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^3\left(C_1 L_1 L_L R_3 g_m r_o + C_1 L_1 L_L R_3\right) + s\left(L_L R_3 g_m r_o + L_L R_3\right)}{R_3 g_m r_o + R_3 + s^4\left(C_1 C_L L_1 L_L R_3 g_m r_o + C_1 C_L L_1 L_L R_3\right) + s^3\left(C_1 C_L L_L R_3 r_o + C_1 L_1 L_L g_m r_o + C_1 L_1 L_L\right) + s^2\left(C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3 + C_1 L_L R_3 + C_1 L_L r_o + C_L L_L R_3 g_m r_o + C_L L_L R_3\right) + s\left(C_1 R_3 r_o + L_L g_m r_o + L_L\right)}$$

9.142 X-INVALID-ORDER-142 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^4\left(C_1 C_L L_1 L_L R_3 g_m r_o + C_1 C_L L_1 L_L R_3\right) + s^2\left(C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3 + C_L L_L R_3 g_m r_o + C_L L_L R_3\right)}{g_m r_o + s^4\left(C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L\right) + s^3\left(C_1 C_L L_1 R_3 g_m r_o + C_1 C_L L_1 R_3 + C_1 C_L L_L R_3 + C_1 C_L L_L r_o\right) + s^2\left(C_1 C_L R_3 r_o + C_1 L_1 g_m r_o + C_1 L_1 + C_L L_L g_m r_o + C_L L_L\right) + s\left(C_1 R_3 + C_1 r_o + C_L R_3 g_m r_o + C_L R_3\right) + 1}$$

9.143 X-INVALID-ORDER-143 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2\left(C_1 L_1 R_L g_m r_o + C_1 L_1 R_L\right)}{g_m r_o + s^3\left(C_1 C_3 L_1 R_L g_m r_o + C_1 C_3 L_1 R_L\right) + s^2\left(C_1 C_3 R_L r_o + C_1 L_1 g_m r_o + C_1 L_1\right) + s\left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L\right) + 1}$$

9.144 X-INVALID-ORDER-144 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^2\left(C_1 L_1 g_m r_o + C_1 L_1\right) + 1}{s^3\left(C_1 C_3 L_1 g_m r_o + C_1 C_3 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1\right) + s^2\left(C_1 C_3 r_o + C_1 C_L r_o\right) + s\left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.145 X-INVALID-ORDER-145 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2\left(C_1 L_1 R_L g_m r_o + C_1 L_1 R_L\right)}{g_m r_o + s^3\left(C_1 C_3 L_1 R_L g_m r_o + C_1 C_3 L_1 R_L + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L\right) + s^2\left(C_1 C_3 R_L r_o + C_1 C_L R_L r_o + C_1 L_1 g_m r_o + C_1 L_1\right) + s\left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L + C_L R_L g_m r_o + C_L R_L\right) + 1}$$

9.146 X-INVALID-ORDER-146 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^3\left(C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L\right) + s^2\left(C_1 L_1 g_m r_o + C_1 L_1\right) + s\left(C_L R_L g_m r_o + C_L R_L\right) + 1}{s^4\left(C_1 C_3 C_L L_1 R_L g_m r_o + C_1 C_3 C_L L_1 R_L\right) + s^3\left(C_1 C_3 C_L R_L r_o + C_1 C_3 L_1 g_m r_o + C_1 C_3 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1\right) + s^2\left(C_1 C_3 r_o + C_1 C_L R_L + C_1 C_L r_o + C_3 C_L R_L g_m r_o + C_3 C_L R_L\right) + s\left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.147 X-INVALID-ORDER-147 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_L g_m r_o + C_1 L_1 L_L) + s (L_L g_m r_o + L_L)}{C_1 r_o s + g_m r_o + s^4 (C_1 C_3 L_1 L_L g_m r_o + C_1 C_3 L_1 L_L + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (C_1 C_3 L_L r_o + C_1 C_L L_L r_o) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_1 L_L + C_3 L_L g_m r_o + C_3 L_L + C_L L_L g_m r_o + C_L L_L) + 1}$$

9.148 X-INVALID-ORDER-148 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_L L_L g_m r_o + C_L L_L) + 1}{C_1 C_3 C_L L_L r_o s^4 + s^5 (C_1 C_3 C_L L_1 L_L g_m r_o + C_1 C_3 C_L L_1 L_L) + s^3 (C_1 C_3 L_1 g_m r_o + C_1 C_3 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_1 C_L L_L + C_3 C_L L_L g_m r_o + C_3 C_L L_L) + s^2 (C_1 C_3 r_o + C_1 C_L r_o) + s (C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L)}$$

9.149 X-INVALID-ORDER-149 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L + s^2 (C_1 L_1 R_3 R_L g_m r_o + C_1 L_1 R_3 R_L)}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s^3 (C_1 C_3 L_1 R_3 R_L g_m r_o + C_1 C_3 L_1 R_3 R_L) + s^2 (C_1 C_3 R_3 R_L r_o + C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3 + C_1 L_1 R_L g_m r_o + C_1 L_1 R_L) + s (C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3 R_3 R_L)}$$

9.150 X-INVALID-ORDER-150 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^2 (C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3)}{g_m r_o + s^3 (C_1 C_3 L_1 R_3 g_m r_o + C_1 C_3 L_1 R_3 + C_1 C_L L_1 R_3 g_m r_o + C_1 C_L L_1 R_3) + s^2 (C_1 C_3 R_3 r_o + C_1 C_L R_3 r_o + C_1 L_1 g_m r_o + C_1 L_1) + s (C_1 R_3 + C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3 + C_L R_3 g_m r_o + C_L R_3) + 1}$$

9.151 X-INVALID-ORDER-151 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_3 R_L g_m r_o + R_3 R_L + s^2 (C_1 L_1 R_3 R_L g_m r_o + C_1 L_1 R_3 R_L)}{R_3 g_m r_o + R_3 + R_L g_m r_o + R_L + s^3 (C_1 C_3 L_1 R_3 R_L g_m r_o + C_1 C_3 L_1 R_3 R_L + C_1 C_L L_1 R_3 R_L g_m r_o + C_1 C_L L_1 R_3 R_L) + s^2 (C_1 C_3 R_3 R_L r_o + C_1 C_L R_3 R_L r_o + C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3 + C_1 L_1 R_L g_m r_o + C_1 L_1 R_L) + s (C_1 R_3 R_L + C_1 R_3 r_o + C_1 R_L r_o + C_3 R_3 R_L g_m r_o + C_3}$$

9.152 X-INVALID-ORDER-152 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^3 (C_1 C_L L_1 R_3 R_L g_m r_o + C_1 C_L L_1 R_3 R_L) + s^2 (C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3) + s (C_L R_3 R_L g_m r_o + C_L R_3 R_L)}{g_m r_o + s^4 (C_1 C_3 C_L L_1 R_3 R_L g_m r_o + C_1 C_3 C_L L_1 R_3 R_L) + s^3 (C_1 C_3 C_L R_3 R_L r_o + C_1 C_3 L_1 R_3 g_m r_o + C_1 C_3 L_1 R_3 + C_1 C_L L_1 R_3 g_m r_o + C_1 C_L L_1 R_3 + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 C_3 R_3 r_o + C_1 C_L R_3 R_L + C_1 C_L R_3 r_o + C_1 C_L R_L r_o + C_1 L_1 g_m r_o + C_1 L_1 + C}$$

9.153 X-INVALID-ORDER-153 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_L R_3 g_m r_o + C_1 L_1 L_L R_3) + s (L_L R_3 g_m r_o + L_L R_3)}{R_3 g_m r_o + R_3 + s^4 (C_1 C_3 L_1 L_L R_3 g_m r_o + C_1 C_3 L_1 L_L R_3 + C_1 C_L L_1 L_L R_3 g_m r_o + C_1 C_L L_1 L_L R_3) + s^3 (C_1 C_3 L_L R_3 r_o + C_1 C_L L_L R_3 r_o + C_1 L_1 L_L g_m r_o + C_1 L_1 L_L) + s^2 (C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3 + C_1 L_L R_3 + C_1 L_L r_o + C_3 L_L R_3 g_m r_o + C_3 L_L R_3 + C_L L_L R_3 g_m r_o + C_L}$$

9.154 X-INVALID-ORDER-154 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s+1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_3 g_m r_o + R_3 + s^4 (C_1 C_L L_1 L_L R_3 g_m r_o + C_1 C_L L_1 L_L R_3) + s^2 (C_1 L_1 R_3 g_m r_o + C_1 L_1 R_3 + C_L L_L R_3 g_m r_o + C_L L_L R_3)}{g_m r_o + s^5 (C_1 C_3 C_L L_1 L_L R_3 g_m r_o + C_1 C_3 C_L L_1 L_L R_3) + s^4 (C_1 C_3 C_L L_L R_3 r_o + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (C_1 C_3 L_1 R_3 g_m r_o + C_1 C_3 L_1 R_3 + C_1 C_L L_1 R_3 g_m r_o + C_1 C_L L_1 R_3 + C_1 C_L L_L R_3 + C_1 C_L L_L r_o + C_3 C_L L_L R_3 g_m r_o + C_3 C_L L_L R_3) + s^2 (C_1 C_3 R_3 r_o +}$$

9.155 X-INVALID-ORDER-155 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_3 L_1 R_3 R_L g_m r_o + C_1 C_3 L_1 R_3 R_L) + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L) + s (C_3 R_3 R_L g_m r_o + C_3 R_3 R_L)}{g_m r_o + s^3 (C_1 C_3 L_1 R_3 g_m r_o + C_1 C_3 L_1 R_3 + C_1 C_3 L_1 R_L g_m r_o + C_1 C_3 L_1 R_L) + s^2 (C_1 C_3 R_3 R_L + C_1 C_3 R_3 r_o + C_1 C_3 R_L r_o + C_1 L_1 g_m r_o + C_1 L_1) + s (C_1 R_L + C_1 r_o + C_3 R_3 g_m r_o + C_3 R_3 + C_3 R_L g_m r_o + C_3 R_L) + 1}$$

9.156 X-INVALID-ORDER-156 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_3 L_1 R_3 g_m r_o + C_1 C_3 L_1 R_3) + s^2 (C_1 L_1 g_m r_o + C_1 L_1) + s (C_3 R_3 g_m r_o + C_3 R_3) + 1}{s^4 (C_1 C_3 C_L L_1 R_3 g_m r_o + C_1 C_3 C_L L_1 R_3) + s^3 (C_1 C_3 C_L R_3 r_o + C_1 C_3 L_1 g_m r_o + C_1 C_3 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (C_1 C_3 R_3 + C_1 C_3 r_o + C_1 C_L r_o + C_3 C_L R_3 g_m r_o + C_3 C_L R_3) + s (C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L)}$$

9.157 X-INVALID-ORDER-157 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1C_3L_1R_3R_Lg_mr_o + C_1C_3L_1R_3R_L) + s^2 (C_1L_1R_Lg_mr_o + C_1L_1R_L) + s (C_3R_3R_Lg_mr_o + C_3R_3R_L)}{g_mr_o + s^4 (C_1C_3C_LL_1R_3R_Lg_mr_o + C_1C_3C_LL_1R_3R_L) + s^3 (C_1C_3C_LR_3R_Lr_o + C_1C_3L_1R_3g_mr_o + C_1C_3L_1R_3 + C_1C_3L_1R_Lg_mr_o + C_1C_3L_1R_L + C_1C_LL_1R_Lg_mr_o + C_1C_LL_1R_L) + s^2 (C_1C_3R_3R_L + C_1C_3R_3r_o + C_1C_3R_Lr_o + C_1C_LR_Lr_o + C_1L_1g_mr_o + C_1L_1 + C}$$

9.158 X-INVALID-ORDER-158 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_mr_o + s^4 (C_1C_3C_LL_1R_3R_Lg_mr_o + C_1C_3C_LL_1R_3R_L) + s^3 (C_1C_3L_1R_3g_mr_o + C_1C_3L_1R_3 + C_1C_LL_1R_Lg_mr_o + C_1C_LL_1R_L) + s^2 (C_1L_1g_mr_o + C_1L_1 + C_3C_LR_3R_Lg_mr_o + C_3C_LR_3R_L) + s (C_3R_3g_mr_o + C_3R_3 + C}{s^4 (C_1C_3C_LL_1R_3g_mr_o + C_1C_3C_LL_1R_3 + C_1C_3C_LL_1R_Lg_mr_o + C_1C_3C_LL_1R_L) + s^3 (C_1C_3C_LR_3R_L + C_1C_3C_LR_3r_o + C_1C_3C_LR_Lr_o + C_1C_3L_1g_mr_o + C_1C_3L_1 + C_1C_LL_1g_mr_o + C_1C_LL_1) + s^2 (C_1C_3R_3 + C_1C_3r_o + C_1C_LR_L + C_1C_Lr_o + C_3C_LR_3g_mr_o + C_3C_LR}$$

9.159 X-INVALID-ORDER-159 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^4 (C_1C_3L_1L_LR_3g_mr_o + C_1C_3L_1L_LR_3) + s^3 (C_1L_1L_Lg_mr_o + C_1L_1L_L) + s^2 (C_3L_LR_3g_mr_o + C_3L_LR_3) + s (L_Lg_mr_o + L_L)}{g_mr_o + s^5 (C_1C_3C_LL_1L_LR_3g_mr_o + C_1C_3C_LL_1L_LR_3) + s^4 (C_1C_3C_LL_LR_3r_o + C_1C_3L_1L_Lg_mr_o + C_1C_3L_1L_L + C_1C_LL_1L_Lg_mr_o + C_1C_LL_1L_L) + s^3 (C_1C_3L_1R_3g_mr_o + C_1C_3L_1R_3 + C_1C_3L_LR_3 + C_1C_3L_Lr_o + C_1C_LL_Lr_o + C_3C_LL_LR_3g_mr_o + C_3C_LL_LR_3) + s^2 (}$$

9.160 X-INVALID-ORDER-160 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_mr_o + s^5 (C_1C_3C_LL_1L_LR_3g_mr_o + C_1C_3C_LL_1L_LR_3) + s^4 (C_1C_LL_1L_Lg_mr_o + C_1C_LL_1L_L) + s^3 (C_1C_3L_1R_3g_mr_o + C_1C_3L_1R_3 + C_3C_LL_LR_3g_mr_o + C_3C_LL_LR_3) + s^2 (C_1L_1g_mr_o + C_1L_1 + C_LL_Lg_mr_o + C_LL_L) +}{s^5 (C_1C_3C_LL_1L_Lg_mr_o + C_1C_3C_LL_1L_L) + s^4 (C_1C_3C_LL_1R_3g_mr_o + C_1C_3C_LL_1R_3 + C_1C_3C_LL_LR_3 + C_1C_3C_LL_Lr_o) + s^3 (C_1C_3C_LR_3r_o + C_1C_3L_1g_mr_o + C_1C_3L_1 + C_1C_LL_1g_mr_o + C_1C_LL_1 + C_1C_LL_L + C_3C_LL_Lg_mr_o + C_3C_LL_L) + s^2 (C_1C_3R_3 + C_1C_3r_o + C_1}$$

9.161 X-INVALID-ORDER-161 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{s^3 (C_1L_1L_3R_Lg_mr_o + C_1L_1L_3R_L) + s (L_3R_Lg_mr_o + L_3R_L)}{R_Lg_mr_o + R_L + s^4 (C_1C_3L_1L_3R_Lg_mr_o + C_1C_3L_1L_3R_L) + s^3 (C_1C_3L_3R_Lr_o + C_1L_1L_3g_mr_o + C_1L_1L_3) + s^2 (C_1L_1R_Lg_mr_o + C_1L_1R_L + C_1L_3R_L + C_1L_3r_o + C_3L_3R_Lg_mr_o + C_3L_3R_L) + s (C_1R_Lr_o + L_3g_mr_o + L_3)}$$

9.162 X-INVALID-ORDER-162 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^3 (C_1L_1L_3g_mr_o + C_1L_1L_3) + s (L_3g_mr_o + L_3)}{C_1r_os + g_mr_o + s^4 (C_1C_3L_1L_3g_mr_o + C_1C_3L_1L_3 + C_1C_LL_1L_3g_mr_o + C_1C_LL_1L_3) + s^3 (C_1C_3L_3r_o + C_1C_LL_3r_o) + s^2 (C_1L_1g_mr_o + C_1L_1 + C_1L_3 + C_3L_3g_mr_o + C_3L_3 + C_LL_3g_mr_o + C_LL_3) + 1}$$

9.163 X-INVALID-ORDER-163 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s+1}\right)$

$$H(s) = \frac{s^3 (C_1L_1L_3R_Lg_mr_o + C_1L_1L_3R_L) + s (L_3R_Lg_mr_o + L_3R_L)}{R_Lg_mr_o + R_L + s^4 (C_1C_3L_1L_3R_Lg_mr_o + C_1C_3L_1L_3R_L + C_1C_LL_1L_3R_Lg_mr_o + C_1C_LL_1L_3R_L) + s^3 (C_1C_3L_3R_Lr_o + C_1C_LL_3R_Lr_o + C_1L_1L_3g_mr_o + C_1L_1L_3) + s^2 (C_1L_1R_Lg_mr_o + C_1L_1R_L + C_1L_3R_L + C_1L_3r_o + C_3L_3R_Lg_mr_o + C_3L_3R_L + C_LL_3R_Lg_mr_o + C}$$

9.164 X-INVALID-ORDER-164 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^4 (C_1C_LL_1L_3R_Lg_mr_o + C_1C_LL_1L_3R_L) + s^3 (C_1L_1L_3g_mr_o + C_1L_1L_3) + s^2 (C_LL_3R_Lg_mr_o + C_LL_3R_L) + s (L_3g_mr_o + L_3)}{g_mr_o + s^5 (C_1C_3C_LL_1L_3R_Lg_mr_o + C_1C_3C_LL_1L_3R_L) + s^4 (C_1C_3C_LL_3R_Lr_o + C_1C_3L_1L_3g_mr_o + C_1C_3L_1L_3 + C_1C_LL_1L_3g_mr_o + C_1C_LL_1L_3) + s^3 (C_1C_3L_3r_o + C_1C_LL_1R_Lg_mr_o + C_1C_LL_1R_L + C_1C_LL_3R_L + C_1C_LL_3r_o + C_3C_LL_3R_Lg_mr_o + C_3C_LL_3R_L) + s^2 (}$$

9.165 X-INVALID-ORDER-165 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2+1}\right)$

$$H(s) = \frac{s^3 (C_1L_1L_3L_Lg_mr_o + C_1L_1L_3L_L) + s (L_3L_Lg_mr_o + L_3L_L)}{L_3g_mr_o + L_3 + L_Lg_mr_o + L_L + s^4 (C_1C_3L_1L_3L_Lg_mr_o + C_1C_3L_1L_3L_L + C_1C_LL_1L_3L_Lg_mr_o + C_1C_LL_1L_3L_L) + s^3 (C_1C_3L_3L_Lr_o + C_1C_LL_3L_Lr_o) + s^2 (C_1L_1L_3g_mr_o + C_1L_1L_3 + C_1L_1L_Lg_mr_o + C_1L_1L_L + C_1L_3L_L + C_3L_3L_Lg_mr_o + C_3L_3L_L + C_LL_3L_Lg_mr_o +}$$

9.166 X-INVALID-ORDER-166 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \frac{L_3 s}{C_3 L_3 s^2+1},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{s^5 (C_1C_LL_1L_3L_Lg_mr_o + C_1C_LL_1L_3L_L) + s^3 (C_1L_1L_3g_mr_o + C_1L_1L_3 + C_LL_3L_Lg_mr_o + C_LL_3L_L) + s (L_3g_mr_o + L_3)}{C_1C_3C_LL_3L_Lr_os^5 + C_1r_os + g_mr_o + s^6 (C_1C_3C_LL_1L_3L_Lg_mr_o + C_1C_3C_LL_1L_3L_L) + s^4 (C_1C_3L_1L_3g_mr_o + C_1C_3L_1L_3 + C_1C_LL_1L_3g_mr_o + C_1C_LL_1L_3 + C_1C_LL_1L_Lg_mr_o + C_1C_LL_1L_L + C_1C_LL_3L_L + C_3C_LL_3L_Lg_mr_o + C_3C_LL_3L_L) + s^3 (C_1C_3L_3r_o + C_1C_LL_3}$$

9.167 X-INVALID-ORDER-167 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^4 \left(C_1 C_3 L_1 L_3 R_L g_m r_o + C_1 C_3 L_1 L_3 R_L\right) + s^2 \left(C_1 L_1 R_L g_m r_o + C_1 L_1 R_L + C_3 L_3 R_L g_m r_o + C_3 L_3 R_L\right)}{g_m r_o + s^4 \left(C_1 C_3 L_1 L_3 g_m r_o + C_1 C_3 L_1 L_3\right) + s^3 \left(C_1 C_3 L_1 R_L g_m r_o + C_1 C_3 L_1 R_L + C_1 C_3 L_3 R_L + C_1 C_3 L_3 r_o\right) + s^2 \left(C_1 C_3 R_L r_o + C_1 L_1 g_m r_o + C_1 L_1 + C_3 L_3 g_m r_o + C_3 L_3\right) + s \left(C_1 R_L + C_1 r_o + C_3 R_L g_m r_o + C_3 R_L\right) + 1}$$

9.168 X-INVALID-ORDER-168 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^4 \left(C_1 C_3 L_1 L_3 g_m r_o + C_1 C_3 L_1 L_3\right) + s^2 \left(C_1 L_1 g_m r_o + C_1 L_1 + C_3 L_3 g_m r_o + C_3 L_3\right) + 1}{C_1 C_3 C_L L_3 r_o s^4 + s^5 \left(C_1 C_3 C_L L_1 L_3 g_m r_o + C_1 C_3 C_L L_1 L_3\right) + s^3 \left(C_1 C_3 L_1 g_m r_o + C_1 C_3 L_1 + C_1 C_3 L_3 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_3 C_L L_3 g_m r_o + C_3 C_L L_3\right) + s^2 \left(C_1 C_3 r_o + C_1 C_L r_o\right) + s \left(C_1 + C_3 g_m r_o + C_3 + C_L g_m r_o + C_L\right)}$$

9.169 X-INVALID-ORDER-169 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^4 \left(C_1 C_3 L_1 L_3 R_L g_m r_o + C_1 C_3 L_1 L_3 R_L\right) + s^2 \left(C_1 L_1 R_L g_m r_o + C_1 L_1 R_L + C_3 L_3 R_L g_m r_o + C_3 L_3 R_L\right)}{g_m r_o + s^5 \left(C_1 C_3 C_L L_1 L_3 R_L g_m r_o + C_1 C_3 C_L L_1 L_3 R_L\right) + s^4 \left(C_1 C_3 C_L L_3 R_L r_o + C_1 C_3 L_1 L_3 g_m r_o + C_1 C_3 L_1 L_3\right) + s^3 \left(C_1 C_3 L_1 R_L g_m r_o + C_1 C_3 L_1 R_L + C_1 C_3 L_3 R_L + C_1 C_3 L_3 r_o + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L + C_3 C_L L_3 R_L g_m r_o + C_3 C_L L_3 R_L\right) + s^2 \left(C_1 C_3 R_L r_o + \cdots\right.}$$

9.170 X-INVALID-ORDER-170 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^5 \left(C_1 C_3 C_L L_1 L_3 R_L g_m r_o + C_1 C_3 C_L L_1 L_3 R_L\right) + s^4 \left(C_1 C_3 L_1 L_3 g_m r_o + C_1 C_3 L_1 L_3\right) + s^3 \left(C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L + C_3 C_L L_3 R_L g_m r_o + C_3 C_L L_3 R_L\right) + s^2 \left(C_1 L_1 g_m r_o + C_1 L_1 + C_3 L_3 g_m r_o + C_3 L_3\right) + s \cdots}{s^5 \left(C_1 C_3 C_L L_1 L_3 g_m r_o + C_1 C_3 C_L L_1 L_3\right) + s^4 \left(C_1 C_3 C_L L_1 R_L g_m r_o + C_1 C_3 C_L L_1 R_L + C_1 C_3 C_L L_3 R_L + C_1 C_3 C_L L_3 r_o\right) + s^3 \left(C_1 C_3 C_L R_L r_o + C_1 C_3 L_1 g_m r_o + C_1 C_3 L_1 + C_1 C_3 L_3 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_3 C_L L_3 g_m r_o + C_3 C_L L_3\right) + s^2 \left(C_1 C_3 r_o + C_1 C_L R_L + C_1 C \cdots\right.}$$

9.171 X-INVALID-ORDER-171 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{s^5 \left(C_1 C_3 L_1 L_3 L_L g_m r_o + C_1 C_3 L_1 L_3 L_L\right) + s^3 \left(C_1 L_1 L_L g_m r_o + C_1 L_1 L_L + C_3 L_3 L_L g_m r_o + C_3 L_3 L_L\right) + s \left(L_L g_m r_o + L_L\right)}{C_1 C_3 C_L L_3 L_L r_o s^5 + C_1 r_o s + g_m r_o + s^6 \left(C_1 C_3 C_L L_1 L_3 L_L g_m r_o + C_1 C_3 C_L L_1 L_3 L_L\right) + s^4 \left(C_1 C_3 L_1 L_3 g_m r_o + C_1 C_3 L_1 L_3 + C_1 C_3 L_1 L_L g_m r_o + C_1 C_3 L_1 L_L + C_1 C_3 L_3 L_L + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L + C_3 C_L L_3 L_L g_m r_o + C_3 C_L L_3 L_L\right) + s^3 \left(C_1 C_3 L_3 r_o + C_1 C_3 L_L \cdots\right.}$$

9.172 X-INVALID-ORDER-172 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ L_3 s + \frac{1}{C_3 s},\ \infty,\ \infty,\ L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{g_m r_o + s^6 \left(C_1 C_3 C_L L_1 L_3 L_L g_m r_o + C_1 C_3 C_L L_1 L_3 L_L\right) + s^4 \left(C_1 C_3 L_1 L_3 g_m r_o + C_1 C_3 L_1 L_3 + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L + C_3 C_L L_3 L_L g_m r_o + C_3 C_L L_3 L_L\right) + s^2 \left(C_1 L_1 g_m r_o + C_1 L_1 + C_3 L_3 g_m r_o + C_3 L_3 + C_L \cdots\right.}{s^5 \left(C_1 C_3 C_L L_1 L_3 g_m r_o + C_1 C_3 C_L L_1 L_3 + C_1 C_3 C_L L_1 L_L g_m r_o + C_1 C_3 C_L L_1 L_L + C_1 C_3 C_L L_3 L_L\right) + s^4 \left(C_1 C_3 C_L L_3 r_o + C_1 C_3 C_L L_L r_o\right) + s^3 \left(C_1 C_3 L_1 g_m r_o + C_1 C_3 L_1 + C_1 C_3 L_3 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_1 C_L L_L + C_3 C_L L_3 g_m r_o + C_3 C_L L_3 + C_3 C_L L_L g_m r_o + C_3 \cdots\right.}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 \left(C_3 C_L R_1 R_3 R_L g_m r_o + C_3 C_L R_1 R_3 R_L\right) + s \left(C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_L R_1 R_L g_m r_o + C_L R_1 R_L\right)}{s^2 \left(C_3 C_L R_1 R_3 g_m r_o + C_3 C_L R_1 R_3 + C_3 C_L R_1 R_L g_m r_o + C_3 C_L R_1 R_L + C_3 C_L R_3 R_L + C_3 C_L R_3 r_o + C_3 C_L R_L r_o\right) + s \left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_L} R_1 R_3 g_m r_o \sqrt{\frac{1}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 R_3 \sqrt{\frac{1}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 R_L g_m r_o \sqrt{\frac{1}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 R_L \sqrt{\frac{1}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_3}\sqrt{C_L} \cdots}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o}$

wo: $\sqrt{\frac{1}{C_3 C_L R_1 R_3 g_m r_o + C_3 C_L R_1 R_3 + C_3 C_L R_1 R_L g_m r_o + C_3 C_L R_1 R_L + C_3 C_L R_3 R_L + C_3 C_L R_3 r_o + C_3 C_L R_L r_o}}$

bandwidth: $\frac{\left(C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o\right) \sqrt{\frac{1}{C_3 C_L R_1 R_3 g_m r_o + C_3 C_L R_1 R_3 + C_3 C_L R_1 R_L g_m r_o + C_3 C_L \cdots}}}{\sqrt{C_3}\sqrt{C_L} R_1 R_3 g_m r_o \sqrt{\frac{1}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 R_3 \sqrt{\frac{1}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 R_L g_m r_o \sqrt{\frac{1}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}} + \sqrt{C_3}\sqrt{C_L} R_1 R_L \sqrt{\frac{1}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}} \cdots}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: $\frac{R_1 R_3 R_L g_m r_o + R_1 R_3 R_L}{R_1 R_3 g_m r_o + R_1 R_3 + R_1 R_L g_m r_o + R_1 R_L + R_3 R_L + R_3 r_o + R_L r_o}$

K-BP: $\frac{C_3 R_1 R_3 g_m r_o + C_3 R_1 R_3 + C_L R_1 R_L g_m r_o + C_L R_1 R_L}{C_3 R_1 g_m r_o + C_3 R_1 + C_3 R_3 + C_3 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o}$

Qz: None

Wz: $\frac{1}{\sqrt{C_3}\sqrt{C_L}\sqrt{R_3}\sqrt{R_L}}$

10.2 X-INVALID-WZ-2 $Z(s)=\left(R_1+\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ R_L+\frac{1}{C_L s}\right)$

$$H(s)=\frac{R_3g_mr_o+R_3+s^2\left(C_1C_LR_1R_3R_Lg_mr_o+C_1C_LR_1R_3R_L\right)+s\left(C_1R_1R_3g_mr_o+C_1R_1R_3+C_LR_3R_Lg_mr_o+C_LR_3R_L\right)}{g_mr_o+s^2\left(C_1C_LR_1R_3g_mr_o+C_1C_LR_1R_3+C_1C_LR_1R_Lg_mr_o+C_1C_LR_1R_L+C_1C_LR_3R_L+C_1C_LR_3r_o+C_1C_LR_Lr_o\right)+s\left(C_1R_1g_mr_o+C_1R_1+C_1R_3+C_1r_o+C_LR_3g_mr_o+C_LR_3+C_LR_Lg_mr_o+C_LR_L\right)+1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}R_1R_3g_mr_o\sqrt{\frac{g_mr_o}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_L}R_1R_3\sqrt{\frac{g_mr_o}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_L}R_1R_Lg_mr_o\sqrt{\frac{g_m \cdots}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_m \cdots}}}{\cdots}$

wo: $\sqrt{\frac{g_mr_o+1}{C_1C_LR_1R_3g_mr_o+C_1C_LR_1R_3+C_1C_LR_1R_Lg_mr_o+C_1C_LR_1R_L+C_1C_LR_3R_L+C_1C_LR_3r_o+C_1C_LR_Lr_o}}$

bandwidth: $\frac{\cdots}{\sqrt{C_1}\sqrt{C_L}R_1R_3g_mr_o\sqrt{\frac{g_mr_o}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_L}R_1R_3\sqrt{\frac{g_mr_o}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_L}R_1R_Lg_mr_o\sqrt{\frac{\cdots}{R_1R_3g_mr_o+R_1 \cdots}}}$

K-LP: R_3

K-HP: $\frac{R_1R_3R_Lg_mr_o+R_1R_3R_L}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}$

K-BP: $\frac{C_1R_1R_3g_mr_o+C_1R_1R_3+C_LR_3R_Lg_mr_o+C_LR_3R_L}{C_1R_1g_mr_o+C_1R_1+C_1R_3+C_1r_o+C_LR_3g_mr_o+C_LR_3+C_LR_Lg_mr_o+C_LR_L}$

Qz: None

Wz: $\frac{1}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_L}}$

10.3 X-INVALID-WZ-3 $Z(s)=\left(R_1+\frac{1}{C_1 s},\ \infty,\ R_3+\frac{1}{C_3 s},\ \infty,\ \infty,\ R_L\right)$

$$H(s)=\frac{R_Lg_mr_o+R_L+s^2\left(C_1C_3R_1R_3R_Lg_mr_o+C_1C_3R_1R_3R_L\right)+s\left(C_1R_1R_Lg_mr_o+C_1R_1R_L+C_3R_3R_Lg_mr_o+C_3R_3R_L\right)}{g_mr_o+s^2\left(C_1C_3R_1R_3g_mr_o+C_1C_3R_1R_3+C_1C_3R_1R_Lg_mr_o+C_1C_3R_1R_L+C_1C_3R_3R_L+C_1C_3R_3r_o+C_1C_3R_Lr_o\right)+s\left(C_1R_1g_mr_o+C_1R_1+C_1R_L+C_1r_o+C_3R_3g_mr_o+C_3R_3+C_3R_Lg_mr_o+C_3R_L\right)+1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}R_1R_3g_mr_o\sqrt{\frac{g_mr_o}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}R_1R_3\sqrt{\frac{g_mr_o}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}R_1R_Lg_mr_o\sqrt{\frac{g_m \cdots}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr \cdots}}}{\cdots}$

wo: $\sqrt{\frac{g_mr_o+1}{C_1C_3R_1R_3g_mr_o+C_1C_3R_1R_3+C_1C_3R_1R_Lg_mr_o+C_1C_3R_1R_L+C_1C_3R_3R_L+C_1C_3R_3r_o+C_1C_3R_Lr_o}}$

bandwidth: $\frac{\cdots}{\sqrt{C_1}\sqrt{C_3}R_1R_3g_mr_o\sqrt{\frac{g_mr_o}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}R_1R_3\sqrt{\frac{g_mr_o}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}+\frac{1}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}}+\sqrt{C_1}\sqrt{C_3}R_1R_Lg_mr_o\sqrt{\frac{\cdots}{R_1R_3g_mr_o+R_1R \cdots}}}$

K-LP: R_L

K-HP: $\frac{R_1R_3R_Lg_mr_o+R_1R_3R_L}{R_1R_3g_mr_o+R_1R_3+R_1R_Lg_mr_o+R_1R_L+R_3R_L+R_3r_o+R_Lr_o}$

K-BP: $\frac{C_1R_1R_Lg_mr_o+C_1R_1R_L+C_3R_3R_Lg_mr_o+C_3R_3R_L}{C_1R_1g_mr_o+C_1R_1+C_1R_L+C_1r_o+C_3R_3g_mr_o+C_3R_3+C_3R_Lg_mr_o+C_3R_L}$

Qz: None

Wz: $\frac{1}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}}$

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